

An Extensional Scientific Realism (I)

The Continuity of Extensions

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Abstract

This is the first of two papers giving a response to the pessimistic meta-induction argument against scientific realism. This paper argues for, and defends the importance of, the continuity of reference. It does so through two related proposals: the first is an application of classical referential semantics, including a mixed account of reference. The second proposal is to make the domain of application of a scientific theory explicit by noticing that extensions are determined by intensions, circumstances of evaluation, and context of utterance. As a result, the extensions are restricted to the theory's domain of application, and continuity can be construed as a matter of extensions. The resulting position is a modest scientific realism, according to which one is justified in believing what confirmed theories say about extensions but not, in general, about intensions. I dub it 'extensional scientific realism'. I answer various objections, and discuss how the position improves on a number of previous realist accounts. The second paper focusses on the notion of approximate truth, and on the differences with structural realism.

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1 Introduction

This is the first of two papers that aim to develop an answer to the pessimistic meta-induction argument against scientific realism. The answer emphasises three main ideas:

- (i) the application of *classical referential semantics* to scientific theories,
- (ii) the explicit specification of the *domain of application* of a scientific theory, and;
- (iii) the development of a qualitative notion of *approximate truth*.

While the two papers focus on answering the pessimistic meta-induction argument, the overall aim is to offer what I take to be a credible scientific realist position which I will call ‘extensional scientific realism’.¹

The position is a relatively straightforward application, to scientific theories, of the distinction between extensions and intensions, and it can be stated in simple terms: *We are justified in being scientific realists about extensions but not, in general, about intensions*. This is inspired by the logical empiricists’ reply to anti-reductionist charges (due to Kuhn and Feyerabend) that meanings change radically when theories change. The logical empiricists’ reply was to admit that intensions *partly* change: but to argue that extensions are *continuous* across theory change.²

While I am not wedded to the cogency of the empiricists’ reply to the anti-*reductionist* challenge, I will argue that one can base on it a good reply to the anti-*realist* challenge. Thus my aim here is not to revive the logical empiricists’ doctrines—since I am interested in developing a *realist* account of science. Rather, I propose that, with appropriate modifications, Nagel’s basic reply to Feyerabend—that, despite the discontinuity of the intensions, the extensions are continuous—is true, and deserves development as a reply to anti-realism.

The view requires a nuanced construal of ‘extension’, that I will develop in conversation with two objections (see below), and that uses (ii) above: the explicit specification of the domain of application of a scientific theory. With these tools in place, my realism is a modest one: I accept the truth of a *partial* discontinuity thesis for intensions, while I argue for the *continuity* of extensions, where the latter are restricted to the theory’s domain of application.

As I announced above, making this reply into a cogent scientific realism requires answering two objections:

(1) To argue that the continuity of the extensions is rich enough that extensional scientific realism does not collapse into either empiricism or into other forms of scientific realism, especially entity realism and structural realism.

(2) To explain why the scientific realist need not worry about allegedly “radical” descriptive differences between theories.

My reply to the first objection involves arguing against what I shall call the ‘simple view’ of reference, according to which the continuity of reference (which contrasts with the large discontinuities of intensions) commits the realist to only a domain of “faceless” individuals, devoid of natures, that are “given merely existentially”. The simple view is sometimes prompted by an erroneous construal of the causal theories of reference.

¹For a recent review of the debate over scientific realism, see Psillos (2018: pp. 22-32).

²See Scheffler (1967: pp. 60-64) and Nagel (1979: pp. 914-915). For the relation between realism and logical empiricism (in terms of the ‘emergent realist tendencies in logical empiricism’), see Neuber (2018).

The response to the simple view proceeds in two related steps that develop idea (i) above:

(I) I will adopt a *mixed theory of reference*, which combines descriptive and causal (I will say: ‘practical’) aspects. On such a theory, the referents satisfy appropriate descriptions, and so they are not faceless: we know more about them than only that they exist.

(II) The mixed theory is in part motivated by the general fact that extensions are determined by *intensions, circumstances of evaluation, and context of utterance*. And thus, the extensions have not just practical, but also conceptual and formal-predictive, aspects.

The steps (I) and (II), together, secure the result, (1), an intuitively appealing version of scientific realism: with a commitment to both observable and unobservable entities, whose natures are described extensionally.

The response (2) involves further consideration of the domain of application of a scientific theory, i.e. idea (ii) above. Namely, well-confirmed scientific theories come, both in philosophy and in science, equipped with limited domains of application. This means that the scientific realist’s commitments to the theory’s extension, thus restricted by the theory’s domain of application, can be justified: but a commitment to the intensions, which involve possible worlds beyond the theory’s domain of application, is not in general justified. Thus in judging the alleged “discontinuities” between successive theories, it is crucial to distinguish a theory’s being applied within its domain of application from outside of it. I will argue that, while one should require theories with significantly overlapping domains of application to be continuous on the overlapping extensions, nothing requires that they agree outside. And thus, the alleged “discontinuity” outside a theory’s domain of application is not really a problem for scientific realism, because there is no expectation in the first place that a theory should be successful in any way, outside its domain of application.

After Kripke (1972) and Putnam (1973, 1975), one popular line of response to the pessimistic meta-induction argument was to replace the (implicitly assumed) descriptive theory of reference by a causal-historical theory.³ The reference of now-abandoned terms can then be retained, despite the changes in the descriptions.⁴ The causal-historical aspect is often combined with the descriptive aspect, thus giving a mixed, causal descriptive theory (see Lewis (1984) and Psillos (1999, 2012)).

But continuity of reference still does not fully justify scientific realism, since many of the now-abandoned terms were once thought to refer. Thus a second principle is often added: namely, the realist commitment should be restricted to only a selected set of entities. Indeed, Kitcher (1993) and Psillos (1999) point out that a scientific theory is not a single monolithic entity that can be an object of belief: for it consists of different kinds of statements and laws, models of various sorts, experimental practices, etc., which should receive various degrees of confidence.

For example, Psillos (1999: p. 107) proposes that one should distinguish between

³The pessimistic meta-induction argument was first formulated by Putnam (1978: p. 25), and given a firm historical basis by Laudan (1981).

⁴For a critical discussion of pure causal-historical theories of reference, see for example Psillos (1999: pp. 274-277).

core and idle elements of theories: so that the theoretical elements which contributed to the success of past theories turn out to be those core elements which were retained in subsequent theories, while those that ‘were not essential for the success of the theory were treated with suspicion’.

Stanford (2006: pp. 153) has pointed out (rightly, I think) that past scientists held many beliefs, such as the belief in the aether, that they thought were core parts of their theories, but which nevertheless were not retained by later theories. Thus a more conceptual distinction is needed: one that applies more homogeneously to a theory.

The distinction between extensions and intensions allows us to say that, at any given moment in time, and in so far as a scientific theory is empirically successful and the scientists have good reasons to use the various terms they use, the terms of a theory refer extensionally, and *homogeneously, for all the terms of the theory* (where, by ‘homogeneous’, I will mean that in general one does not need to distinguish between different theory parts or between entities). And to respond to the pessimistic meta-induction argument, which is concerned with the problem of the referential continuity of different theories, we will say that there is an extensional equivalence between well-confirmed theories one of which succeeds the another, even though some (even many!) of their terms are intensionally inequivalent: and that this extensional equivalence is sufficient to secure the conceptual continuity that is disputed by the pessimistic meta-induction argument.

A number of authors, including Stein (1989), Papineau (2010), Stanford (2015), and Myrvold (2020), have proposed abandoning the question of the reference of terms altogether, and to focus on the substantial issues instead: namely, ‘getting clear about what we do, and do not, agree about with our predecessors’ (Myrvold, 2020: p. 2). I will agree with these authors that an alleged “failure of reference” is not what matters, and that verdicts about the agreement or disagreement between theories should not be based just on contingent historical choices about the terms that scientists use to talk about entities.

However, I will also argue that altogether dismissing the topic of reference—especially, the insights that classical semantics and theories of reference can give us—would be throwing the baby out with the bath-water. The important question, for the kind of scientific realism that I will defend, is whether we are justified in believing that things are, substantially, the way well-confirmed scientific theories say they are, on a literal but cautious reading. And here, semantics, including the analysis of reference, surely helps.

Section 2 summarises the discussion of the pessimistic meta-induction argument, and sketches my contribution to it. Section 3 then applies classical referential semantics to scientific theories and defines extensional equivalence, i.e. the notion that gives the desired continuity between theories. Section 4 analyses extensional equivalence in terms of a number of correspondence relations. Section 5 then compares with one version of entity realism, and discusses an objection by Stanford. Section 6 is the conclusion.

The companion paper De Haro (2020a), which I shall refer to as (Part II), further develops extensional scientific realism by introducing a qualitative notion of approximate truth, i.e. idea (iii) above. It also compares scientific realism with structural realism.

2 Scientific Realism and its Critiques

My motto in this Section is: listen to your enemy. Or, in more equable terms: take the criticisms of scientific realism seriously, and make them a starting point for scientific realism. Section 2.1 first gives a general characterisation of scientific realism. Section 2.2 then summarises the pessimistic meta-induction argument. I will agree with some aspects of this critique. Section 2.3 then sketches a version of scientific realism that is free of these problems.

2.1 Which scientific realism?

In this paper I defend a version of scientific realism that is committed to scientific theories that give an empirically adequate description of some domain of application. Roughly speaking, scientific realism can be characterised as follows:

Scientific realism is the belief in the approximate truth of scientific theories that are empirically well-confirmed in a (significant) domain of application.

This general characterisation of scientific realism is still imprecise, since the various notions need to be unpacked. It is also potentially vulnerable to the *pessimistic meta-induction argument*, which starts by noting that past scientific theories that were well-confirmed in a significant domain of application have turned out to be false. And since the past well-confirmed, now discarded, theories were false, *belief in the truth* of today's well-confirmed theories is unjustified.

Before I go on to describe the pessimistic meta-induction argument in more detail, let me motivate a bit more my general characterisation of scientific realism. Scientific realism is often characterised, as I did above, in terms of assertions and beliefs about the *epistemic status of scientific theories*, i.e. in terms of approximate truth (see e.g. Chakravartty (2017: Section 1.1)). However, it is sometimes alternatively characterised in terms of the epistemic *aims* of science: famously, by van Fraassen (1980: p. 8)—who, in addition, introduces a philosophical word, viz. ‘scientific gnostic’, for the person who believes the science she accepts to be true (1994: p. 182).

Of course, no one wants to call themselves a ‘gnostic’ (and the word, as defined by van Fraassen, has to do only with belief, and not with justification): but names and puns apart, there are several reasons to resist the replacement of talk about beliefs by talk about aims. Mine are as follows: First, ‘aim’ is too weak a term, and a definition of scientific realism in terms only of the aims of science, unless strengthened by some other condition, is liable to requiring the realist to be a realist about activities that have the same aims as science, but are not science. Second, the aims of science are themselves a subject of philosophical debate, and it does not seem desirable to make the definition of scientific realism directly dependent on the outcome of that debate. Third—and this is my main reason—scientific realism is *philosophical realism about something specific*: namely, scientific theories, and in particular the descriptions that they give of the world (for the moment, I use ‘description’ here in the broad sense, not always necessarily linguistic). To be a realist about some range of entities, in the general philosophical sense, is to believe in the mind-independent *existence* of those entities: it is not to have any specific beliefs about our *aims* concerning those objects. Thus it would be a mistake to turn the

definition of scientific *realism* into a definition of the *nature* of science. Scientific realism is an important foundational question in the philosophy of science, but it is not identical to the question ‘What is science?’

2.2 The pessimistic meta-induction argument and its responses

In his vigorous attack on scientific realism, Laudan (1981) *severs the link between the ‘success’ of theories* (i.e. the degree to which theories are confirmed) *and their reference*, i.e. the entities that the terms of theories refer to. Thus he criticizes the idea that ‘the world probably contains entities very like those postulated by our most successful theories’ (p. 22), which one would expect to be true, if well-confirmed theories are true or approximately true.⁵

Indeed, he famously lists a number of theories that allegedly were once successful but whose central terms for unobservable entities we now regard as “non-referring”, including (p. 33): the electromagnetic aether; the caloric theory of heat; the crystalline spheres of ancient and medieval cosmology; the phlogiston theory of chemistry; etc.

Laudan’s charge that the old theories ‘fail to refer’ or, more accurately, that the referents of old theories and of their successors are very different, goes back of course to Kuhn and Feyerabend’s ideas about meaning and incommensurability. Feyerabend (1963: pp. 929-931, 939-941) emphasises that *meanings* change across theory boundaries, because the meaning of every term depends upon the theoretical environment in which it occurs.

The problem of reference that Laudan points to has been said to be particularly acute for *descriptive theories of reference*, according to which the terms for unobservable entities refer in virtue of how these entities are described by the theory. This is indeed the problem of meaning change stressed by Kuhn and Feyerabend. Thus if a theory is overturned, in general its successor will describe the unobservable entities very differently to how the old theory did, and the referents of the old theory and of its successor will often be different. There is then no reason why even our best current theories will fare better than previous theories did. In short: on a descriptive theory of reference, if the descriptions change, the referents also change.

Kripke’s (1972) and Putnam’s (1973, 1975) *causal-historical theory* of the reference of proper names and other terms allowed for a novel way to address the pessimistic meta-induction argument, because reference is fixed and transmitted through causal and historical mechanisms, rather than using descriptions.⁶

Applied to natural science, the causal theory of reference says that a term for an unobservable entity refers to whatever entities cause the phenomena that led theorists to introduce the term in the first place. So, ‘when Franklin and Ampère used the term ‘electricity’, it referred to the *actual* electricity causing the phenomena (like sparks, lightning

⁵Laudan is particularly concerned with the realist’s claim that realism explains why science is successful (the ‘no-miracles argument’). That is, he questions the cogency of the realist’s abductive argument, of inference to the best explanation. In this paper, I will set aside the no-miracles argument and questions of explanation and confirmation (for a recent review, see Saatsi (2018) and Wray (2018)).

⁶Psillos (2012: pp. 214-221) gives a brief overview of the history of the descriptive and causal theories of reference.

bolts, currents, and electromagnets) that led them to posit the existence of an unobservable physical magnitude responsible for these phenomena' (Stanford, 2006: p. 148). In the same way, the 'aether' of the early electromagnetic theories refers to the electromagnetic field itself, because it plays the same sorts of causal roles that the electromagnetic field is supposed to play in carrying electromagnetic energy.

The causal theory of reference has been said to make reference too easy, and thus that it is liable to ascribing reference to words like 'aether', 'caloric', 'phlogiston', etc. that we no longer think refer to anything.

Causal descriptivism was introduced by Lewis⁷ (1984: pp. 226-227) as a 'halfway house' between descriptivism and causal theories of reference. I will call such theories, in general, *mixed theories of reference*. On Psillos' (1999: p. 277; 2012: p. 222) version of causal descriptivism, the reference of some term to an entity is fixed when this entity both causes the phenomena of interest, and satisfies a causal description of how it causes those phenomena. The causal contact anchors the descriptions to the referent; the descriptions enable that the referent is tracked across different situations, and secure that reference does not come too easily.⁸

The causal descriptivist theory of reference can be used to develop a *selective scientific realism*, which commits the realist only to the belief in the parts or terms of the theory that are causally involved or, alternatively, only in those occasions in which they are causally involved, in producing the phenomena. For example, Kitcher (1993) proposes that we distinguish the tokens of theoretical terms that refer on certain occasions, from those tokens that do not. His motivation (1993: Chapter 4), apart from the criticism of purely descriptive theories of reference, is that a scientific term can refer in different ways on various occasions of use. Thus the speaker's *intentions* must be considered in assigning reference: 'thus Priestley's tokens of 'dephlogisticated air' are fixed by his intention to refer to air with the substance emitted in combustion removed from it' (p. 149). This allows Kitcher to say that some tokens of Priestley's 'phlogiston' refer, while others do not.

There is a historical objection to selective realist theories of this type: despite Kitcher's explicit consideration of speaker intentions, the theory is nevertheless liable to the objection that it 'radically disconnects what a theorist is talking about from what she thinks she is talking about' (Ladyman and Ross, 2007: p. 87). Namely, although Priestley was using the single word 'phlogiston' in a variety of occasions, we now, with the benefit of hindsight, judge—arguably *against* his intentions—that he was "really" using the same word to refer to different entities. (I will come back to this objection in Section 5.2).

A further line of response to the problem of reference has been to state that 'this concern for reference... is a distraction from what really matters' (Stein, 1989: p. 57). Stein has argued, in particular, that, apart from terms that were abandoned, there are other terms, like 'atom', that are retained, but change their meaning, in later theories.⁹

⁷Lewis introduces causal descriptivism as part of a response to Putnam's (1976, 1981) model-theoretic argument that the truth-values of the sentences of a theory can be made true by any model of the right cardinality, however unintuitive its interpretation. For a discussion, see Button (2013: pp. 14-17, 21-22).

⁸For a critique of Psillos's (2012) mixed theory of reference, see Hoefer and Martí (2020: pp. 17-20).

⁹This is of course familiar from Feyerabend's critique of meaning, which focussed on the different meanings of 'mass', in special relativity and in classical mechanics. And that the word 'planet' changed meaning

In other words, whether a term appears in the current scientific vocabulary or not is a fairly inconsequential matter, subject to various historical contingencies, and a defence of scientific realism should not focus on it.

Papineau (2010: pp. 376-377) has argued that, while descriptivist theories are ‘stingy’ (because reference easily fails) and causal theories are ‘generous’ (because far more terms successfully refer), the scientific realist should not set their hopes on mixed theories of reference either, because the success of scientific realism should not hinge on a theory of reference that delicately fine-tunes these two aspects. While Papineau does not discard the possibility that some theory of reference might be right, he argues that this does not matter, because reference is irrelevant to approximate truth. He proposes a new variation on the Ramsey sentence that weakens the assumptions involved in the descriptions used to fix the causes of the phenomena, so that a unique cause can be identified without committing to reference. Furthermore, talk about approximate truth can be stated in terms of this modified Ramsey sentence directly. For example, different choices of the assumptions of the more restricted description have varying degrees of approximate truth.¹⁰

Stanford (2015 pp. 1-3) also endorses Stein’s (1989) point that reference is not the notion that matters for the debate over scientific realism (as does Myrvold (2020)).¹¹ He defends a historicist view, according to which the real question is whether:

... the future of science will exhibit the same broad pattern of repeated, profound, and unpredictable changes in fundamental theoretical orthodoxy that historicist critics of scientific realism argue characterizes its past... [S]uccessful scientific accounts of various parts of nature are repeatedly replaced with even more impressive and powerful successors making fundamentally different claims about the constitution and/or operation of those parts of nature.

Further, Stanford (2015: p. 26) urges that the most fundamental problem with disputed terms is not whether they refer or not, but

... that they have not ultimately turned out to be part of the most pragmatically powerful and successful conceptual apparatus we have for thinking and talking about the phenomena to which they (arguably) *do* refer.

Even though the history of terms *is* important, it is easy to agree with Stein (1989) that it would be superficial for philosophers to focus only on, and to draw directly ontological conclusions from, the historical continuity of the use of *words*: in so far as this may be a terminological and historically highly contingent process, not reflective of what the

during the scientific revolution is of course an example of incommensurability in Kuhn (1962: p. 115).

¹⁰If a scientific theory T has a high degree of confirmation, a Ramseyfied version of it will have a high degree of approximate truth, regardless whether T ’s terms refer to anything. Papineau proposes the Ramsey-type formula $(E\Phi)[T_r(\Phi) \wedge (\Phi)(T_r(\Phi) \rightarrow \Phi = \Psi) \wedge T(\Phi)]$, where $T(\Phi)$ is the theory involving the term Φ , and T_r is a restricted version of the theory T , i.e. a subset of its assumptions or descriptions. The restriction to T_r undercuts the holism involved in determining a term Φ from the entire theory $T(\Phi)$. Thus $T_r(\Phi)$ includes the subset of assumptions of T that descriptively fix Φ (and leaves out some claims of T that may be dubious). The above formula says that there is a unique term Φ , satisfying the restricted description $T_r(\Phi)$, and such that $T(\Phi)$ is also satisfied.

¹¹That “reference is irrelevant” is certainly a controversial thesis: for recent work focussing on the question of reference, see for example Hoefer and Martí (2020).

philosopher might wish to say about ontology. It is also easy to agree with these authors that the insistence on the single notion of “failure of reference” is a red herring. Indeed, some of the language used in these debates, for example the metaphorical phrases “fixing the reference” and “transmission of reference”, can easily mislead us into thinking that the act of securing reference is like capturing a horse with a lasso—you either succeed, so that you have it at the other end of your rope, or fail. And if you bet your money on that horse, you make sure to track it on its race, through the colours that the horse and its rider are wearing.

Misguided as some of the language used in the debates about reference may be, we should not get distracted by that language. Too often in philosophy, the rhetoric of a debate leads us to think that we have been all wrong about what the debate was about, that it is really about something else than we thought it was, and that we should really abandon a concept since it no longer does us any service: thereby often producing the pendulum to swing to the exact opposite side (using a metaphor by Psillos (2012: pp. 214-229)). Therefore, it would be a mistake to simply accept the way in which Stein and Stanford frame the debate about scientific realism, i.e. it would be too easy to dismiss the problem of reference altogether.

Let me explain this in more detail. The view that ‘reference is not the question’ appears to be motivated, at least in part, by what I shall call the ‘simple view’ of reference. (I certainly can think of no other serious motivation to abandon reference than one that appeals to the simple view, which I will now explain.)

The definition, in the philosophy of language and logic, of the extension of a sentence as its truth-value, could mislead us into thinking that the referent of the collection of sentences of a theory is ‘merely’ a collection of truth values, devoid of content.¹² This impression can be reinforced by the standard definition of the extension of a property as a set of individuals (or, alternatively, as a characteristic function from a set of individuals to a set of truth values), and of the referent of a name as an individual. Thus that reference is contentless (i.e. a set of truth values), and that the individuals in question are faceless and “given merely existentially”—that our task as philosophers is to make sure that we “anchor” our words to them, and “track” them throughout history, even if the reference itself, i.e. the initial rope with the lasso during capture, and the horse’s colours during the race, do not tell us almost anything about the individual, which is “simply there”. This caricature of reference is what I call the ‘simple view’. More generally, the simple view is any view that makes reference into something poorer and more simplistic than it actually is.¹³ If something like the simple view were true, I would agree that the question can be safely dropped altogether.

¹²For example, Button (2013: pp. 14-17) discusses Putnam’s (1981: pp. 32-38) permutation argument purely in terms of collections of ‘given objects’, without regard for the intensions. But that permutations can be applied even though intensions are assigned to names, is an important part of Putnam’s argument (p. 35). For all his merits, Button’s first four chapters discuss problems of reference and correspondence without any mention of intensions. Although this might be for pedagogical reasons, such pedagogy seems to further entrench the simple view.

¹³The simple view has its roots, so far as I know, in Fine (1975: p. 19), in a simplistic reading of Putnam’s (1973, 1975) causal theory of reference, and in the implicit assimilation, by Feyerabend (1963: pp. 929-931, 939-941), of ‘meaning’ with ‘intension’. For a discussion of this, see Section 5.1.

But the simple view is mistaken, because extensions are of course partly determined by intensions, and so they are not contentless or faceless at all. Furthermore, despite the potentially misleading talk about “fixing”, “anchoring”, and “tracking”, it would be uncharitable—and, in my opinion, plainly incorrect—to take authors like Kitcher and Psillos to be defending some shallow notion of reference, i.e. some version of the simple view.

When philosophers of science have asked ‘whether a term refers’, they were often not asking how one should check the theory’s list of truth values, but rather a deeper question: hence the often-used phrase whether terms ‘genuinely refer’ (or that a term is ‘substantially non-referring’). They were not necessarily denying that reference is successful in the way that our use of words is successful in daily life, but rather that the reference relation in question satisfies some standard that is appropriate to scientific theories and to the question of scientific realism.

Indeed, reference is a qualitative matter, since most of our words will, most times, refer to *something*, even if they perhaps describe it using inappropriate terms (see Lewis (1984: pp. 227-228)).

Also, the literature’s (sometimes criticised) focus on single terms should be seen as a placeholder for, or a first step towards analysing, sentences and entire theories. Once we move from the reference of single terms to the truth of sentences and theories, what initially appeared to be a shallow focus becomes much more relevant. The focus on single terms is a step required to address the theory’s approximate truth.

Thus I think that Stein and Stanford on the one hand, and Kitcher and Psillos on the other, agree more about what the debate is about than its dynamics and rhetoric might suggest.¹⁴

I have argued that we should not make ‘reference’, as in the simple view, into something more shallow and impoverished than it is. The question about the reference of terms is not primarily whether the entities *named* by the theory exist, but whether these entities are substantially *as the theory says they are*. Although Laudan’s initial question about reference was partly formulated in terms of ‘whether various entities (like the aether, phlogiston, etc.) exist’, the question was not *only* about existence, but also about whether we are justified in believing that entities then named ‘aether’, ‘atom’, ‘caloric, etc. can be identified with entities that we now name using different, or with the same, words—as the elaborate replies to the pessimistic meta-induction argument also attest. Thus the main question for the realist has always been about the *continuity* of reference: whether there is a continuity of reference that is strong enough to justify the scientific realist’s talk of ‘approximate truth’.

2.3 A sketch of the overall proposal

Section 2.3.1 first sketches the two main points of this paper. Section 2.3.2 then answers three objections.

¹⁴For example, Stanford’s (2015) criticism on this point appears to be addressed not primarily to realists, but to his fellow anti-realists. His criticism of Kitcher is also not about the point here under scrutiny, but about Kitcher’s apparent neglect of interpretative choices when assessing reference in historical cases.

2.3.1 Two main points: reference and domains of application

This paper can be summarised in points (i) and (ii) of Section 1: the first is about reference in classical semantics, and the second is about scientific theories. Although these points are not new, they have, so far as I know, not been used systematically before to formulate an answer to the pessimistic meta-induction argument, and to develop a scientific realist view.

The point about reference is simple: we should model questions about reference—understood, as explained in the previous Section, as the broad question about the continuity of the descriptions given by previous scientists with those of our best current theories—using classical referential semantics. In classical referential semantics, extensions are determined by intensions, circumstances of evaluation, and context of utterance (for more details, see Section 3).

This leads us naturally to consider a *mixed theory of reference*, where the intensions are typically general descriptions, the context of utterance includes aspects such as ostension, and the circumstances of evaluation are descriptive, but include aspects (such as times, places, etc.) that would normally be part of a causal-historical theory.¹⁵

This referential semantics can be straightforwardly applied to scientific theories, and thus, with appropriate adaptations, used to assess the continuity of reference between different theories. More precisely, the question will not just be ‘whether the theory’s terms refer to anything’, but whether the similarities between the referents of successive historical theories, that are well-confirmed and are about the same topic, is strong enough that we can say that their referents are the same. I will introduce the term *extensional equivalence* to denote such similarity, and I will analyse extensional equivalence into descriptive and practical factors (Section 4).

Second, I will point out something that is obvious but seems too often neglected in discussions of scientific realism: the fact that scientific theories have limited domains of applicability. Setting aside for the moment the possibility of a final theory, ordinary scientific theories have limited domains of validity, and they can always be applied to possible worlds, or situations, where the theory turns out to be false. That is, the set of worlds where a theory can be applied is larger than the set of worlds where it is true. Even restricting to the actual world, no scientific theory is actually proposed as a description of *all* phenomena, and no theory is supposed to describe all phenomena, present and future, in full accuracy. This is why we talk about domains of phenomena where theories apply.

Also, no *prudent* scientist introduces a scientific theory as an actually accurate description of the phenomena in domains far beyond the domain where the theory has been tested. These domains are limited along different dimensions: not only in terms of the *kinds* of phenomena that are produced, but also in terms of the *conditions* under which

¹⁵It is worth mentioning some differences with Lewis’ (1984: p. 227) and Psillos’ (2012: p. 222) causal descriptivism, where the descriptions are causal or largely causal. While in my case some descriptions will indeed be couched in causal terms, there is no requirement that they should be always, or largely, couched in such terms. Also, Psillos does not seem to take into account how the domain of application, including its measurement and calculational accuracies, is delimited (at least, not explicitly). On the other hand, Lewis (p. 228) *does* make clear that reference ‘is a matter of degree, making total theory come true is a matter of degree, the two desiderata trade off... The terms of trade are vague; that will make for moderate indeterminacy of reference; but the sensible realist won’t demand perfect determinacy’.

those phenomena are produced.

This is an almost obvious point: when specifying a scientific theory, we should also specify its purported domain of application, i.e. a part of the actual world. Future experiments may allow us to enlarge that domain, or may tell us that the theory fails beyond it. But if a theory is mature enough, i.e. it is well-confirmed, its applicability within its domain becomes fairly certain (and of course in empirical science, as in epistemology, we cannot, and do not, aim for absolute certainty, but only to develop justified, true beliefs).

For example, Newtonian physics could be used very well, for several centuries, to make sufficiently accurate navigation calendars: and we do not really expect it to be overturned in that domain. Also, given certain conditions within our galaxy, general relativity can be used very well to predict the motions of the planets in the solar system, and we do not really expect it to be overturned in that domain; and so on.

What does this imply for our realist commitments? The proposal is of course that we should trust scientific theories *only* in their domains of application, i.e. in the domains where they have been well-tested. It will be useful to restate this in the language of extensions and intensions, as follows.

When we consider the *intension* of a sentence, we need to consider all the possible worlds to which the theory applies. But this includes many worlds that are different from the actual world, and parts of the actual world where the theory has not been well-tested, and thus are outside the theory's domain of application, because we have no good reason to believe that the theory will be true there (even though we might hope, and might even have heuristic arguments to the effect, that it will be). Thus there are possible worlds that look nothing like the actual world but where the intensions (of the sentences of a well-tested theory) are true, and in general there are parts of the actual world where they are *false*. And so, a cautious scientific realist should in general not be a realist about intensions. Indeed, we should not be realists about the theory's alleged *propositions* considered in the abstract, i.e. separate from the theory's domain of application. Rather, we should be realists about the extensional sentences derived from those propositions, i.e. *under appropriate circumstances of evaluation and context of utterance*. In other words, we should only be realists about *extensions*, restricted to the theory's domain of application.

This is deeply engrained in the practice of physics, even if it is not always emphasised in philosophical accounts of theories. A scientific theory is not a collection of universally applicable theorems, but rather comes with a specification of the kinds of situations to which the theory is supposed to apply. For example, general relativity is expected to break down if the typical length scale of the system approaches the Planck length (and this is something that we can say even without having a theory of quantum gravity that replaces general relativity). And in effective quantum field theories, a cut-off is often imposed beyond which the theory requires modification or even replacement. Of course, one may be lucky and find, in some cases, that the theory is still valid beyond the envisaged domain: but physics is hard, and one should not in general expect such luck.

Thus my proposal is that *we are justified in being scientific realists about extensions but not, in general, about intensions*. Here, it is essential that the extensions are not understood as in the simple view, and also that they should be restricted, through the appropriate circumstances of evaluation and context of utterance, to the theory's domain

of application in the actual world.

2.3.2 Answering three objections

In this Section, I will give replies to three objections to the views summarised in Section 2.3.1.

(i) *Anti-realism?* Very well, the anti-realist might say: theories come with domains of application, and you should not commit yourself beyond those... fair enough. But if you cannot commit yourself to intensions but only extensions, then you are effectively defending a form of anti-realism. For you drop the intension, i.e. precisely that which describes those things that the anti-realist looks at with suspicion: ontology, concepts, and meanings. You are left with only existence claims and truth-values, but there is little that you can say about the entities themselves, since you no longer have intensions. And such poor content is no comfort for the scientific realist. At best, you have here a form of empiricism, since you commit yourself only to the theory's predictions about observable entities.

No: for this would be to fall again into the trap of the simple view. I will give two further arguments for this. First, the intensions are not completely dropped by the extensional scientific realist: recall that the intensions are (along with the circumstances of evaluation and the context of utterance) used in determining the extensions. Thus the intensions, though imperfect, clearly do capture correct aspects of the world, i.e. in conjunction with the right circumstances of evaluation and context of utterance they deliver correct extensions. In a slogan: *imperfect intensions, restricted and corrected by appropriate circumstances of evaluation and context of utterance, can deliver correct extensions.*

Second, ontology, concepts, and meanings have both extensional and intensional aspects. This second point can perhaps be made clearest by addressing the above criticism in the objection, that I am defending a form of empiricism. The observable-unobservable distinction of course has nothing to do with the distinction between extensions and intensions (not even on a generous construal of 'observable'). For example, electrons and forces are usually regarded as unobservable entities, and yet they are the extensions of the corresponding terms. In other words, because extensions are partly determined by intensions, they still capture the theory's ontology; just not in all possible worlds, but only under certain circumstances of evaluation and context of utterance that apply to the theory's domain of application.

Thus extensional scientific realism is a modest realism: we recognise (temporarily setting aside the possibility of a final theory) that we are not, in general, justified to believe the theory's intensions regardless of the theory's domain of application, i.e. a part of the actual world.

Here, we should add two points. First, that this view is only natural and not surprising, given the scientific practice: as I mentioned earlier, prudent scientists always present their theories as descriptions of limited domains of reality. Indeed, it would be hubris to expect the intensions of a scientific theory to be true in general, regardless of a domain of application. (Note that the modesty required here is more specific than simply saying that 'it is likely that even our best theories turn out to be false'). And second, that

extensional scientific realism is *realism enough*, because it includes the commitment to that which has always been the bone of contention in the debate about scientific realism: namely, unobservable entities. The view implies that our ontology should be piecemeal and advance step by step, from smaller domains to larger, more inclusive, domains (I will return to this in (Part II)). It would be wrong to abandon ontology altogether, since extensions (when correctly understood, as against the simple view I sketched above) are ontology enough.

The more ambitious side of this statement of extensional scientific realism is that scientific theories, *even if their intensions are imperfect*, still have extensions that are approximately true, and in which we can have a justified belief.

(ii) “*Great descriptive differences*”. Stanford has emphasised the great descriptive differences between old theories and their successors. To make his point clear, he uses rich language (with commendably few repetitions!), including phrases like ‘repeated, profound, and unpredictable changes in fundamental theoretical orthodoxy’ (2015: p. 1), ‘impressive and powerful successors making fundamentally different claims’ (p. 2), ‘repeatedly and reliably mistaken’ (p. 4), ‘radically misconceived’, ‘radically and fundamentally distinct’ (p. 8), ‘substantive change in our beliefs’ (p. 17), ‘massive changes’ (p. 20), ‘profound discontinuities’ (p. 24), ‘fundamental, profound, far-reaching, and unpredictable’ differences (p. 26). From here, it appears that one can pose a very general objection: if the descriptive differences between successive theories are so radical, how does any kind of theoretical continuity that can possibly be identified help the scientific realist?

But in view of my discussion above, I take the analysis assumed by the objection, namely the mention of “great descriptive differences”, to be too coarse: since it does not distinguish between extensions and intensions. Namely, the analysis is not sensitive to two important points:

(1) The fact that extensions are conceptually rich, and suffice for a realist commitment to both observable and unobservable entities.

(2) The recommendation that, for ordinary scientific theories, one should *expect* intensions to fail, and that prudent scientists are very well aware of this (this awareness neither requires nor implies any pessimistic meta-induction). And thus, that there is no reason why one should have expected intensions in general to agree, or be committed to them (independently of circumstances of evaluation and context of utterance), in the first place. The differences in intension that are not tied to appropriate circumstances of evaluation and context of utterance are irrelevant, since the cautious scientific realist is not committed to them anyway.

Thus emphasising the “great differences”, without making these sorts of distinctions, fails to capture something important about how a realist commitment can be justified.

Indeed, this kind of objection is strongly reminiscent of Kuhn and Feyerabend’s claims,¹⁶ which I discussed in Section 2.2, of conceptual differences across theory boundaries leading to radical changes. The changes in the intensions may be as radical and massive as you wish; the changes in the extensions are piecemeal.

Let me summarise. The scientific realist does not possess any justified beliefs about

¹⁶See e.g. Putnam (1973: p. 200).

those far-away domains, outside of the theory’s extension, where the theory has not yet been tested, and where it can be expected to fail. Thus when the theory fails in those other domains outside of its extension, and the realist resorts to a new theory, this does not imply a “radical” change of beliefs, since the scientific realist did not, and could not, possess any justified beliefs about how the theory would fare in a new domain of phenomena in the first place.

Thus the most important constraint that a scientific realist should place on a construal of ‘approximate truth’ is not a purported measure of how close one is to a final theory, but rather the inclusivity of the domains (this will be the topic of (Part II)). As long as the differences between two theories on the old theory’s domain are negligible, relative to the old theory’s level of accuracy, the commitment to the old theory’s extension is preserved. In this sense, the theory was, and still is, approximately true in that domain. In the next Section, I will show that one can use referential semantics to give a criterion of extensional equivalence between terms: and, by a widening of the notion, between theories.

(iii) *Success matters the most.* Does extensional scientific realism satisfy Stanford’s (2015: p. 26) criterion of continuity, or ‘fundamental problem’ about abandoned terms, quoted in Section 2.2? The criterion is whether certain terms have ‘ultimately turned out to be part of the most pragmatically powerful and successful conceptual apparatus we have for thinking and talking about the phenomena to which they (arguably) *do* refer’.

The extensional scientific realist can readily satisfy this criterion (the criterion may not be sufficient for some scientific realists, but no matter). For example, Newtonian mechanics, including its notion of ‘gravitational force’, surely *is* among ‘the most pragmatically powerful and successful conceptual apparatus we have for thinking and talking about’ the phenomena involving the motions of the sun and the moon, and applied in navigation, even after Newtonian gravitation has been superseded by general relativity. Namely, Newtonian mechanics is still today taught in both introductory and advanced courses in universities, not just for pedagogical reasons, but precisely because of its ‘pragmatically powerful and successful conceptual apparatus’ for describing nearby phenomena of gravitation.¹⁷ Within the admissible margins of error, one would be crazy to use the full non-linear theory of general relativity to predict those motions. Indeed, it would be crazy to attack any problem in terrestrial mechanics or in navigation starting from a non-Euclidean spacetime. We still use the notion of the gravitational force, because we know it is—for the kinds of situations under discussion, including our terrestrial frame of reference—a good enough approximation to the truth. That is, whether we use Newtonian mechanics, or general relativity, the relevant extension is the same, and this entity “looks” (it is an unobservable quantity!) like a gravitational force, i.e. both the gravitational force and the spacetime curvature cause the same phenomena on that domain if we restrict to the earth’s system of coordinates. After all, the Newtonian force can be *derived* in general relativity!¹⁸ Indeed, the gravitational force is extensionally equivalent to the spacetime

¹⁷The temporary mention, in this paragraph, of ‘courses taught in universities’, rather than historical science, of course responds to the need to consider mature science, reflected in textbooks: and also to the fact that Stanford’s criterion is about the pragmatic success of those theories *today*, and so it is appropriate to consider *today’s* textbooks.

¹⁸Newtonian gravitation can indeed be derived from general relativity in the weak field approxima-

curvature on that extension.¹⁹

Also Maxwell's theory, in its domain of application, surely *is* one of those successful theories. Indeed, it is no problem that 'aether' is no longer used in the modern presentation of Maxwell's theory. For, if we can show that, under the relevant circumstances of evaluation and context of utterance, 'aether' essentially refers to the same thing as the electromagnetic field (something that I will argue in (Part II), see e.g. Hardin and Rosenberg (1982: pp. 611-614)), then Maxwell's theory and our own version of it are extensionally equivalent. Then it is irrelevant that the word 'aether' is no longer used (as Stanford himself admits!). Indeed, what is relevant is that, in its domain of application, the theory is still pragmatically powerful and has a successful conceptual apparatus, where the old concepts are extensionally equivalent to the new ones.

This is also *in spite of the alleged "radical differences" between the two theories under consideration, outside the domain of application of the older theory*. Thus this shows that large intensional differences are *compatible* with predictive, conceptual, and experimental success in a given domain of application. Psillos (2009: p. 71) has a good slogan for this: *success is one thing, replacement is another*.

Thus I claim that restricting theories to the domains where they are applicable, rather than taking them as collections of sentences that should be applicable in every possible domain of phenomena, shows that most mature, i.e. theoretically well-developed and empirically well-confirmed, theories satisfy the criterion proposed by Stanford. Under a proper understanding of the domain of application, Stanford's fundamental problem is grist to the extensional scientific realist's mill.²⁰

These results are not simply achieved through a liberalization of the notion of 'approximate truth'. As I argued in point (ii)-(2) above, given that scientific theories have limited domains of application, there is no justification for believing them beyond their proper domains, and their failure outside of them is irrelevant to their approximate truth on those domains, and to the question of scientific realism: namely, whether theories are approximately true *on their domain of application*. An ordinary scientific theory is not devised to succeed beyond its domain of application, and we cannot expect nature to be so kind to us as to guarantee the universal validity even of our best theories. If a theory is successful beyond its domain of application, we are simply in luck. As I stressed ear-

tion: see Wald (1984: p. 77). For a more detailed treatment of the 'Newtonian limit', see Malament (1986: p. 198).

¹⁹There is, in this example, evidently also a question of emergence in Newtonian gravitation, compared to general relativity. But the kind of reduction, involved in extensional equivalence, that I am discussing here is not incompatible with emergence, as I will discuss in (Part II). Indeed, I construe ontological emergence as essentially a change in the intension, regardless of whether or not there is a corresponding change in the extension.

²⁰On this point, Stanford (2015: p. 26) gives, as examples from physics, Ptolemy's 'wandering stars' and Priestley's 'dephlogisticated air'. But these are of course controversial examples, about which some realists argue that the theories in question were not mature enough, while others argue that the terms do in fact refer to the same entities as our current theories, at least in some occasions, even though we no longer use the words. In either case, these examples are not really problematic for scientific realism. Using the less controversial examples, as I did in the main text, of Newtonian mechanics and Maxwell's theory shows that, if the desired criterion is that the theories be 'pragmatically powerful and conceptually successful', then these well-confirmed and now, as is sometimes said, "discarded" (or rather, superseded) theories do satisfy this criterion.

lier, it would be hubris to think that a scientific theory must answer absolute questions, regardless of the circumstances of evaluation and the context of utterance. And thus I can see no motivation at all for a construal of ‘approximate truth’ that requires that the theory’s intension to be true.

3 Meaning and Continuity: between Extensions and Intensions

One of the main jobs that referential semantics is supposed to do for the debate over scientific realism that I will review in the next Chapter is to secure the continuity between theories from different historical epochs. Thus this Section will apply some further—indeed, elementary—aspects of classical intensional semantics to scientific theories.

The distinction between *two kinds of meanings* of words when used by scientists, namely intensions and extensions, is standard in referential semantics and well-known in philosophy of science: but it seems nevertheless to have been largely neglected in this discussion:²¹ while it is prominent in the early literature on *reduction*.²² For example, Nagel (1979: pp. 914-915) used that distinction to defend his own account of reduction against Feyerabend.

The extensional scientific realism that I develop here is semantic and epistemic, but not (strongly) metaphysical.²³ That is, I aim to argue that ‘the realist is justified in believing in the entities that well-confirmed scientific theories postulate (i.e. in endorsing their ontologies), where the meaning of the terms is construed extensionally’. But I have no intention to specify what those entities are, from the outset, and independently of the specifics of the relevant scientific theories.

²¹Kitcher (1993: pp. 75-76) does, in one place, mention the distinction between Frege’s terms ‘sense’ and ‘reference’, but then seems to dismiss it, because Frege allegedly believes that ‘sense determines reference’ (i.e. two expressions with the same sense must share the same referent), but “fixing” reference by using sense is problematic. (This criticism is already in Putnam (1975: p. 219)). I have four arguments against such criticisms. (i) I do not use the Fregean contrast between sense and reference, but rather the Carnapian-Lewisian contrast between intension and extension, which is now standard in semantics. See for example Heim and Kratzer (1988: Chapters 2 and 12) and Speaks (2019). For the comparison between the pairs sense-reference vs. intension-extension, see Carnap (1947: pp. 122-133). (ii) In classical referential semantics, it is *not* the case that the intension alone determines the extension. In particular, it is not true that two expressions with the same intension must share the same extension: context of utterance and circumstances of evaluation are in general also required (Kaplan, 1977: p. 494). (Frege (1956) [1918] of course never denied (ii)). (iii) I do not use a descriptive, but a *mixed* account of reference, where a moderate indeterminacy of reference through descriptions is compensated by practical aspects. (iv) The inaccuracy of the intensions is offset against the inaccuracy of the domain of application. About ‘moderate indeterminacy of reference’ and vagueness, see footnote 15. About the evaluation of claims that are vague or indeterminate in content (through the use of a modified Ramsey sentence) see Papineau (2010: p. 384) and footnote 10.

²²The philosopher’s traditional account of reduction is Nagel (1961: pp. 351-363; 1979), Hempel (1966: §8), Schaffner (2012). On this account, reduction is, essentially, deduction of one theory from another, almost always using additional definitions or bridge-principles linking the two theories’ vocabularies. Recent proposed rebuttals of the objections to the traditional account are: Dizadji-Bahmani et al. (2010: pp. 403-410) and Butterfield (2011: §3).

²³See for example Chakravartty (2017).

3.1 Meaning as a matter of intension and extension

This Section introduces the distinction between intensions and extensions that our analysis will need. Then it introduces a notion of extensional equivalence that will play a central role in what follows, and briefly addresses the question of how extensions are determined.

Recall that, quite generally, the *intension* of a *term* is the term’s linguistic meaning (cf. the Fregean sense),²⁴ as described by the theory, and thus relative to all possible worlds described by the theory.²⁵ The term’s *extension* is its worldly reference (cf. the Fregean ‘reference’), relative to a single possible world, i.e. the (set of) thing(s) or entity(ies) being referred to, relative to a given possible world (with all of its contingent details). Thus ‘Evening star’ and ‘Morning star’ have different intensions (i.e. different linguistic meanings, and the stars they refer to are different in some possible worlds), but they have the same extension in our world, viz. the planet Venus—thus it is a contingent fact about our world that, while their intensions differ, their extension is the same.

Let me say a bit more about the ‘worlds’ that are assumed by standard semantic analyses, as the latter apply to scientific theories.²⁶ In philosophy of science, ‘possible worlds’ are often construed as the various instantiations of a theory, or sets of solutions of the theory’s dynamical equations. Some of these instantiations or sets of solutions describe a part of the actual world (for example, a solution of Newtonian gravitational theory will describe the solar system, while another solution will describe a system in another galaxy, even though there might not be a single solution of Newtonian theory that accurately describes both).

By ‘terms’, we mean the individual expressions, be they simple or composite, i.e. bits of scientific language (e.g. mathematical or other technical language) that appear in the theory and that need to be interpreted: for example, words for entities or properties (‘bacteria’, ‘DNA strand’, ‘cell mobility’, etc.), expressions on one side of an equation, mathematical symbols for force, mass, charge, fields, etc.

Standard texts on semantics distinguish between three kinds of linguistic expressions, which Carnap calls ‘designators’: namely, sentences, individual expressions (such as names, descriptions or words for properties²⁷) and predicates (such as words for relations

²⁴The differences between the two pairs of expressions, ‘sense and reference’ vs. ‘intension and extension’, have to do mainly with opaque contexts and with some of Frege’s more specific doctrines (see Carnap (1947: pp. 122-133)). Apart from opaque contexts, the two pairs of expressions are used in the same way, and give the same result, as Carnap (1947: pp. 125-126) explicitly shows. Where they differ, it is the contrast ‘intension vs. extension’ that I endorse.

²⁵For more about intensional semantics, see Carnap (1947: pp. 177-182) and Lewis (1980). For an excellent introduction to intensional semantics in connection with sentence structure, see Heim and Kratzer (1988: Chapters 2 and 12). For an overview of theories of meaning, see Speaks (2019).

²⁶In philosophy of physics, a distinction is sometimes made between ‘kinematically possible worlds’ and ‘dynamically possible worlds’. Since I will take a theory to also include a dynamics, we will be mostly interested in the dynamically possible worlds.

²⁷I am here giving a uniform treatment of names, descriptions, and words for *properties* (i.e. I regard all of them as individual expressions) because scientific theories tend to treat properties (for example, ‘mass’, ‘electric charge’) as abstract entities. Thus, the term ‘mass’ can appear on the left-hand side of an equation, e.g. $M = 50 \text{ kg}$, as if it were an object. In official expositions of semantics, properties are treated in verb phrases (‘is heavy’). It is a grammatical matter of no consequence whether a property is represented by an individual expression, i.e. a singular term, or as a predicate (the two are each others’

and verb phrases). Different kinds of extensions and intensions are assigned to each of these kinds of linguistic expressions. The case just discussed—the referents of ‘terms’, or individual expressions—are indeed the most interesting cases for scientific realism: since we are typically interested in whether terms such as ‘aether’, ‘caloric’, etc. refer (with the qualifications about reference made in Section 2.2). But let me briefly introduce the semantics of the two other kinds of linguistic expressions—

The extension of a *sentence* (for example, a statement of a law, or an equation) is its truth-value, while its intension is the corresponding proposition (alternatively, a map from the set of worlds described by the theory to truth-values). The extension of a *predicator* (for example, the verb phrases ‘never decreases over time’ or ‘is equal to zero’) is a *characteristic function* from a set of individuals to truth values (alternatively, it is the set of individuals itself), while its intension is a function from the set of worlds to its characteristic function (alternatively, the corresponding property). The referents of sentences will be relevant when we discuss truth, in (Part II). For the moment, we focus on individual expressions.

The extensions of terms thus denote “local”, this-worldly items that are contingently related to the term used—contingent in the sense that the term gives that extension only in a specific possible world described by the theory, depending on its circumstantial details. Intensions, on the other hand, denote “global” items, and thus apply to all possible worlds.²⁸

Our main concern in this Section will be with terms that, like ‘Morning star’ and ‘Evening star’, have different intensions but the same extension.²⁹ Furthermore, we will be interested in cases in which the equality is more general: it holds not only in a single world (i.e. instantiation, solution, or model of the theory), but in various possible worlds—although in the next Section we are interested in applying it to parts of the actual world. Thus it will be useful to introduce the following notion:

Extensional equivalence:³⁰ two terms are extensionally equivalent, with respect to a set of possible worlds described by the theory, iff they have the same extension in each of those worlds (even though their intensions may differ). Two terms that are extensionally equivalent are often said to be ‘coextensive’ or to ‘co-refer’.³¹

How does one know that the extensions are the same in a given world? How does one know that ‘Morning star’ and ‘Evening star’ have the same referent, namely Venus? This is, in astronomy as in other sciences, partly a theoretical and partly an empirical question: one works out deductively the empirical consequences of the use of the term in the

duals): and my usage has the advantage just mentioned. In this paper, we are interested in individual expressions and sentences not verb phrases.

²⁸Thus Lewis (1980: pp. 22-27) defines intensions as maps from linguistic terms (or sequences thereof) to extensions.

²⁹As I mentioned at the end of Section 2.2, this is part of the process of considering sentences and entire theories, which I do explicitly in (Part II).

³⁰See also Carnap (1947: pp. 14, 18). An example: Norton (2012: p. 211), without using the word ‘extension’, discusses an example: ‘[t]he “caloric” of caloric theory refers to the same thing as the “heat” of thermodynamics, but in the confines of situations in which there is no interchange of heat and work.’

³¹‘Extensional equivalence’ is, in philosophy, usually used for the actual world only. The notion of ‘co-reference’ is used in the debate about scientific realism in e.g. Fine (1975: p. 27): see Section 4.4.

theory, applied to the particular empirical circumstances, including the experimental and observational accuracy, and sometimes also the context of utterance. The identification of ‘Evening star’ obviously depends on various details: not only does Venus need to then be on one side of the Sun, trailing it in its course; also, the sky needs to be sufficiently clear and dark. More generally, the planet Venus is open to a more detailed investigation through spectroscopic observations, radio echo, and space probes. This will be the topic of Section 4.

I will next argue that extensional equivalence can be expressed using the *language that we use to talk about the world*, but let me first say a few words about this language. We describe the domain of application partly linguistically (e.g. ‘the nucleus of an eukaryotic cell is surrounded by a nuclear envelope that is perforated by nuclear pores’) and express identity in terms of this language. This language is an aid (e.g. words used to mention objects) to describe the intended physical things and properties, and it is in general different from the technical, formal, or mathematical language of the theory. It is ‘the language of (experimental) physicists’ used to describe the world, and it may contain *extra-linguistic elements* such as diagrams, images, ostension, etc.³²

Thus my account will be a mixed account of reference, which stresses the need to combine descriptive and practical aspects of correspondence in order to establish extensional equivalence.³³

3.2 Extensions are determined by intensions, circumstances, and context

In this Section, I will apply the general intensional semantics introduced in Section 3.1 to scientific theories. This Section focusses on the general elements that determine extensions. Section 4 then explains how, once the extensions are determined, we can assess extensional equivalence.

In classical semantics, extensions are determined by intensions and, in addition, context of utterance and circumstances of evaluation. The circumstances and the context are both made up of indices. Roughly speaking, the context of utterance has to do with the *speaker* (or where and when the expression is uttered) while the circumstances have to do with the particular state that is relevant for the expression, given the context. Thus roughly, the distinction is between ‘who speaks and where and when’ vs. ‘what is being

³²Galison uses the metaphor of the ‘trading zone’ to understand how scientists from different traditions engage with each other, developing pidgins or creole languages: ‘people have and exploit an ability to restrict and alter meanings in such a way as to create local senses of terms that speakers of both “parent” languages recognize as intermediate between the two. The resulting pidgin or creole is neither absolutely dependent on nor absolutely independent of global meanings’ (1997: p. 47). For examples of what I have called ‘the language that scientists use to describe the world’ and of incommensurability, see De Haro (2019: p. 6; 2020: pp. 22-23).

³³The aspects of correspondence that establish extensional equivalence are in broad agreement with the three elements that, according to Galison (1997: pp. 799-800), secure the possibility of interaction and communication between different scientists: namely, theory, experiment, and instrumentation: ‘we do not expect to see the abrupt changes of theory, experimentation, and instrumentation occur simultaneously... When a radically new theory is introduced, we would expect experimenters to use their best-established instruments, not their unproven ones.’

said and about what’. Rather than going further into the semantic details here,³⁴ let me just say how I will use these expressions for scientific theories: namely, the relevant contrast will be between ‘the scientists and their activities’ vs. ‘the specific application of the theory and its (set of) models’.

Thus I will use the phrase *circumstantial physical conditions* for the choice of state of the world (see Section 3.1), together with other physical conditions relevant in determining an extension, and which are modelled by, or have a correlate in, the theory or model. These elements are, in theories of reference, usually called ‘descriptive’. They are, for example, boundary conditions and other conditions about the empirical situation considered: including measurement or observational accuracies, the approximations made, in so far as these correspond to physical properties of the system considered (for example, low speed, or large orbital angular momentum, etc.). They have a formal, mathematical, or technical counterpart in the theory, which I will call the theory’s *level of abstraction*.³⁵ The level of abstraction depends on the properties of the states and quantities that are relevant for a specific system, together with the way in which other practical aspects are implemented in a theory or model. Thus the level of abstraction is the formal correlate that reproduces (and thus determines) the circumstantial physical conditions under which the theory is applicable.³⁶ (I will use the phrase ‘the level of abstraction determines the circumstantial physical conditions’ in this sense).

In what follows, I will use the phrase *domain of application* to refer to a possible world (in the usage introduced in Section 3.1) with specific circumstantial physical conditions. Since the circumstantial physical conditions include approximations (see below), this means that a domain of application of a theory, unlike a possible world with a specified state, is usually not an exact solution of the theory’s dynamical equations, but only an approximate solution.³⁷

Let me here summarise, without aiming for completeness, the main content of the various *levels of abstraction* (and the *circumstantial physical conditions* they determine) involved in determining the extension from the intension:

- The specification of a model, viz. the relevant kinematic and dynamical situation

³⁴For an introduction and examples, see Speaks (2019: Sections 2.1.4 and 2.3). See also Kaplan (1977: pp. 494-495), Lewis (1980) and Perry (1979).

³⁵I take term ‘level of abstraction’ from Floridi (2011: Chapter 3). Levels of abstraction have an objective character, in so far as they determine circumstantial physical conditions which belong to a physical system and to which the theory applies.

³⁶Levels of abstraction are in the realm of language, while the circumstantial physical conditions are in the world. (Both are usually expressed linguistically).

³⁷In the discussions of correspondence between two theories, T and T' (or reduction of one to the other), the literature often uses the notation ‘ T^* ’ to indicate an intermediate theory, T^* , that relates T' and T , since they are not *directly* in correspondence (see e.g. Lewis (1970: pp. 441, 445)). I will not use this notation, because it assumes that T' corresponds to T , via T^* , *globally*, i.e. at the level of the theory. But this is of course often not the case, since one often reduces *models* (rather than theories) to models, and each model has its own circumstantial physical conditions, which may be inconsistent with those of some other model. Thus I will compare *domains of application*, which are worlds, each with its own circumstantial physical conditions. Because the circumstantial physical conditions for each of the worlds are already included in the domain of application, I do not need to consider the intermediary T^* ; as I said before, such a theory need not exist.

(including choices of initial and boundary conditions, velocities, energies, distances, assumptions about the population, environmental conditions, chemical composition, etc.).

- The values of the theory’s and the model’s free parameters (such as mass, coupling strength, chemical concentration, etc.).
- Extra-theoretical facts, in so far as these are modelled by, or incorporated into, the theory or model: including the allowed measurement or observational errors; other influences, such as noise, not described by the theory; aspects of the experimental—including material—conditions, weather conditions, etc.
- Approximations and idealisations (both theoretical and empirical) carried out in order to accommodate all of the above, i.e. to make the theory empirically adequate under the stated conditions and accuracies.

The phrase *context (of application)* will denote all those aspects that are compatible with, but are not fully modelled by, the theory or the relevant models, and that bear on how reference is established in practice. This includes what is often called ‘causal’, although I prefer ‘practical’, since it includes scientific methods and practices, the material preparation and composition of samples, experimental procedures, ostension and intentions of speakers, influences (or lack of them) that are not described by the theory, etc.

Scientific practice has been the focus of several prominent accounts of science, especially the “Stanford school”,³⁸ which range from anti-realist to realist views.³⁹ The focus on scientific practice has sometimes been accompanied by a repudiation, or at least a critique, of theory. I will argue that the critics are right in insisting that theory alone is not enough. But this does not militate against scientific realism: for, as I hope to show, there is a sensible scientific realist account that makes use of both theory and practice.

Given a specific level of abstraction, and often also a context of application, the extensions of two theories with very different intensions can be identical. For a given level of abstraction that determines a specific set of physical conditions, the two theories refer to the same physical system.

³⁸See, for example, Galison (1997), who does not easily qualify as either a realist or an anti-realist (he qualifies his own position as a ‘historicized neo-Kantianism’). As can be seen from my earlier quotations of Galison, my account could be seen as adding a semantic/truth dimension to the aspects of theory, experimentation, and instrumentation that he focusses on (as a result of which our accounts of course differ widely).

³⁹Besides the Stanford school, important realist accounts are Cartwright (1983), Hacking (1983), Massimi (2016, 2018), Chang (2012, 2018), and Radder (1991, 2012 [1984]). The accounts of Massimi, Chang, Radder, and mine are, in several aspects, kindred accounts: however, there is much in which they differ. There are, for example, some structural similarities with Massimi’s account (e.g. between her ‘context of use and context of assessment’ (2016: pp. 356-357) and my ‘circumstantial physical conditions and context of application’), but there are many differences in content—more than I can discuss here. Also, my account agrees with Chang (2018: pp. 176, 179) that theories can contradict one another (in some ways) and still be true (in other ways), but does not need to endorse Chang’s pragmatism, or the metaphysics that underpins his pluralism (2018: pp. 181-184). For a comparison with Radder’s (2012 [1984]) ‘referential realism’, see Section 4.2.

4 Extensional equivalence as given by correspondence

The previous Section addressed how the reference of the terms of scientific theories is “fixed”. Thus I argued that extensions are, as in classical semantics, in general determined by the specification of the intensions, circumstances of evaluation, and context of utterance—which in the context of scientific theories I called ‘circumstantial physical conditions and context of application’. While this is not essentially different from how this is done in ordinary language, there are of course aspects that are highly specific to science.

In this Section, I turn to the aspects that are involved in assessing *extensional equivalence* between theories. In the literature, this is sometimes called the ‘tracking’ of reference, although, as I mentioned in Section 2.2, this word can have the misleading connotations of what I call the ‘simple view’. It is the *continuity of reference* between theories that we are assessing, and we do this again using a mixed account of reference with both descriptive and practical aspects.

Since the question here is not ‘how to get from intensions to extensions’ (this was already addressed in the previous Section), but rather ‘how we can compare the extensions of different theories’, i.e. on *already constituted* extensions, it is best to organise the descriptive and practical aspects slightly differently. For although the questions, from the previous Section, about the constitution of the extension (for example, the specification of a model, the values of the theory’s free parameters, etc.) are still relevant, for the purpose of comparing different theories, we need a more abstract view, where we draw conceptual distinctions in how theories are compared, and identify therein a number of *correspondence relations*.

As for the *descriptive* component of extensional equivalence, we will consider three kinds of correspondence: *predictive* correspondence, *conceptual* correspondence, and *descriptions of causal roles* in a theory or model (more generally, relations within a theory or model).

As for the *practical* component of extensional equivalence, we will consider: measurement procedures, experimental manipulation of materials, and the identity of bits of matter stuff; instrumentation and other experimental procedures are in general also relevant.

As I mentioned in Section 2.2, the discussion will here focus on the extensional equivalence of *terms*: but, in doing so, we of course also consider the extensional equivalence of *sentences*, and more generally the correspondence between different *theories*.

In the literature, these inter-theoretic relations go under the generic name of ‘correspondences’. In this Section, we will consider the case in which there is a sufficiently strong relation of *reduction* between the relevant models of the two theories, as applied to a given domain of application: such that the reducing theory essentially subsumes the reduced theory, and the two theories are extensionally equivalent, on the domain of the reduced theory.⁴⁰ This does not rule out the presence of emergence: I will return to emergence in (Part II).⁴¹

⁴⁰For more about reduction, see footnote 22.

⁴¹See also De Haro (2019) and Butterfield (2011).

4.1 On the notion of ‘correspondence’

In the philosophical literature, the notion of correspondence⁴² is present in discussions of reduction, and various weakenings, of it: Nagel’s (1961, 1979) heterogeneous reduction admits that strict deduction of one theory to another fails, but deduction can nevertheless be achieved by postulating additional **bridge laws**. It is a *correspondence* account of reduction, where the bridge laws are correspondence rules as empirical hypotheses relating things mentioned in both the reduced and the reducing theories. These empirical hypotheses determine the extensions of the reduced theory’s predicates mentioned in the correspondence rules.⁴³ Regardless of Nagel’s empiricism, my own view is that determining extensions is not only an empirical, but also a conceptual matter: indeed, this is one of the central aspects of my proposal, as I argued in the previous Section, and in more detail in this Section.

A further step is taken by Fodor (1974), who denies the complete reduction of the special sciences to physics, on the basis that bridge laws are not identity statements between types, but rather contingent identities between tokens: what he calls **token identity** (especially: of events, rather than properties).⁴⁴ I will use the idea of token identity here, divorced of its anti-reductionist connotations.

Token identity can be recognised through various empirical and conceptual procedures, and not only through the strict identity of the bits of matter stuff of e.g. a sample of a substance, i.e. an individual lump. For example, recognising that, in a specific case, mean kinetic energy and temperature are contingently identical is aided by judging the roles

⁴²Bohr used the word ‘correspondence’ to denote various aspects of the relation between classical and quantum theories. Correspondence, as the broad idea that the successor theory should build on, or should somehow reproduce, the theory that it replaces, of course goes back much earlier. It is surely implicit in Maxwell’s kinetic theory of gases no less than in his unification of electric and magnetic phenomena, and in Newton’s derivation of Kepler’s and Galileo’s laws as special cases of his own theory of gravitation. Post (1971) begins with quotes from Whewell, Duhem, Born, Rohrllich, Maxwell, and Wigner that effectively support the idea of correspondence between theories. For a treatment of Bohr’s correspondence principle, see Rynasiewicz (2015); for an alternative treatment, see e.g. Bokulich (2008). See also van Dongen et al. (2020: Section 5.1). For the correspondence between special relativity and classical mechanics, see Brown (1993).

⁴³Nagel (1979: pp. 914-915) distinguishes two types of correspondence rules: those of the first type state conditions, often in terms of a micro-theory, for the occurrence of traits characterising various things, often macroscopic ones; such bridge laws are empirical hypotheses concerning the extensions of the *predicates* mentioned in the correspondence rules. The bridge laws of the second type relates the *individuals* or entities designated by different predicates. As I mentioned in Section 3, this was a response to Feyerabend’s (1963: pp. 929-931, 939-941) accusation that Nagel did not respect meaning variance: but Nagel (1961: p. 357) had already said that, for example, the meaning of ‘temperature’ in thermodynamics and in classical mechanics is different, but that nevertheless a postulate can be introduced that relates them (‘this postulate cannot be warranted by simply explicating the meanings of the expressions contained in it’). In his 1979 reply, Nagel stated explicitly that ‘temperature’ has a different intension in the two theories, but the same extension.

⁴⁴Sober (1999: pp. 552-554) argues that multiple realisability does not preclude reductionism, and criticises one aspect of Fodor’s (1974) anti-reductionist argument: namely, his contention that laws cannot be disjunctive. Regardless of whether or not Sober’s critique invalidates Fodor’s argument: the aspect of Fodor (1974) that is interesting for scientific realism—namely, his consideration of token rather than type relations between theories—is immune to Sober’s critique, and is close to the view of extensional continuity that I am defending.

that temperature and kinetic energy play in the two theories, including their **causal roles** (the heating up of a gas at constant pressure *causes its expansion*: in one case because of the Boyle-Charles ideal gas law, and in the other because of the free motion of the particles. More on ‘causal roles’ below⁴⁵).

Post (1971) construed the (generalised) correspondence principle as ‘the requirement that any acceptable new theory should account for its predecessor by ‘degenerating’ into that theory under those conditions under which [the predecessor theory] has been well-confirmed by tests’ (p. 16).⁴⁶ Hartmann (2002) and Radder (1991) perceptively analyse and criticise the subsequent literature on the generalised correspondence principle. They factorise the generalised correspondence principle into various factors,⁴⁷ i.e. correspondence relations between different aspects of theories, of which the most relevant for our discussion of single terms are: numerical, formal, and conceptual correspondence (the last is also called ‘conceptual continuity’). In **numerical correspondence**, the two theories agree (perhaps after appropriate approximations are done) on the numerical values of a distinctive set of salient quantities. In **formal correspondence**, the (mathematical or technical) form of the laws of one theory can be obtained from those of the other (Radder (1991: p. 208)). Together, numerical and formal correspondence are called **predictive correspondence**.

Radder’s own conclusion is that between classical and quantum mechanics there is numerical and formal correspondence, but no conceptual correspondence. This is an important difference between our accounts. For in the next Section I will argue that, although there is no conceptual *identity* (i.e. intensional equivalence of concepts) there is *conceptual correspondence* (i.e. extensional equivalence of concepts, and in addition similarity of intensions). Although numerical, formal, and conceptual correspondence between theories on a given extension are a matter of equivalence of models, the *circumstantial physical conditions* listed in the previous Section usually play an important role in obtaining this correspondence.

Post’s ‘conditions under which’ the new theory reproduces the predecessor theory are what I call the ‘levels of abstraction’ (see Section 3.2) at which the theory or model are treated, and which determines the circumstantial physical conditions of the domain. Thus, for example, the approximations made to get numerical correspondence between quantum mechanics and classical mechanics are part of the level of abstraction required to describe the domain of classical physics.

4.2 Conceptual correspondence in quantum mechanics

In this Section, I illustrate the notion of **conceptual correspondence** using the example of quantum mechanics.

One might wonder whether there is *conceptual* correspondence at all between quantum and classical mechanics. Indeed, the conceptual *discontinuity* between classical and quantum mechanics is sometimes assumed to be obvious: namely, quantum mechanics

⁴⁵See also Lewis (1972: pp. 249-252).

⁴⁶For an overview and critical discussion of both Post’s work and a collection of essays in his honour, see Hartmann (2002).

⁴⁷Hartmann (2002: pp. 85-87) identifies seven such factors.

introduces a number of concepts that are not identical with any classical concepts (for example, the quantum state, observables as operators, etc.).

But one should recall the difference between intensions and extensions. For I will argue that, in the domain where classical mechanics is applicable, there is sufficient conceptual correspondence between classical and quantum mechanics: because the main concepts in classical mechanics can be reduced, on their *extensions*, to those of quantum mechanics. It does not seem possible to do this on their *intensions*: and herein lies the possibility of emergence.

For example, the concept of ‘position’, in the quantum theory, corresponds to the concept of ‘position’, in the classical theory, in the following sense. They play similar conceptual roles in both theories: namely, they both give information about the spatial location of a system (without, in the quantum theory, defining the position with infinite precision). They play similar roles also in terms of the **relations** they bear with other concepts, for example with the canonically conjugated ‘momentum’, and of how the corresponding quantities are measured (more on ‘conceptual roles within a theory’ below). In specific situations, i.e. in the domain of application of classical mechanics, the classical concept of position can be reduced to that in quantum mechanics.

Indeed, there is a long tradition of studies and theories along the lines of such relations. Two standard examples are: (1) The reduction of classical to quantum mechanics, whereby classical Poisson brackets go over in commutators⁴⁸ (hinted at at the end of the previous paragraph). (2) Ehrenfest’s theorem, whereby the expectation value of an operator satisfies a close analogue of the classical equation of motion, and *reproduces* the latter in the classical limit (for example, the expectation value of the position operator satisfies an analogue of Newton’s equation).⁴⁹

In both cases, the classical result is usually reproduced in the limit $\hbar \rightarrow 0$ and $N \rightarrow \infty$, where $\hbar$ is Planck’s constant and N is, roughly, the size of the system, e.g. the number of particles or components.⁵⁰ Once the limit is taken, we indeed have formal identity of the quantum and classical results.

The correspondence is also conceptual, because the expectation value of the position

⁴⁸The idea of replacing Poisson brackets by commutators goes back to Dirac (2001 [1964]: pp. 35-43, 77); see also Messiah (1969: pp. 318, 337). The idea of a ‘quantisation map’ was first proposed by Heisenberg, who reinterpreted the mathematical symbols of the classical equations of motion in terms of matrices. It was then used by Schrödinger in understanding his own equation: for a lucid discussion, see Landsman (2007: pp. 427-428). For the mathematical implementation of Heisenberg’s idea (of a systematic map between classical and quantum objects) by von Neumann, see (ibid, p. 430). For more modern and mathematically rigorous approaches to this problem, especially ‘deformation quantisation’, see Landsman (ibid, pp. 446-464).

⁴⁹See Messiah (1969: pp. 316-319), Landsman (2007: pp. 475-476).

⁵⁰The limit $\hbar \rightarrow 0$ should be interpreted as a limit of small $\hbar$ *compared to* an action/energy scale usually set by the quantum mechanical potential, the mass of the particle, etc. See Messiah (1969: p. 214). For a thorough discussion, see Landsman (2007: pp. 471-515). This is not to claim outright a solution to the measurement problem—but it is surely relevant to it. See, for example, the recent programme in Landsman (2013: pp. 378-383), who treats classical and quantum mechanics using a common C^* -algebra formalism. He then shows convergence of judiciously chosen quantum states to classical states (and the corresponding expectation values of quantities converge to the classical values). Another approach to the emergence of quasi-classical behaviour is in terms of the dynamics of decoherence, see Joos and Zeh (1985), Joos et al. (2003) and Zurek (2002).

operator plays a similar role to the position of the particle in the classical theory: as we have seen, it obeys Newton's second law; and, after the limit is taken, the quantum and classical quantities can be *identified*.⁵¹ Thus, using bridge laws, these central concepts of classical mechanics reduce to those of quantum mechanics.

Under the consideration of **counterfactual situations** in other possible worlds, this similarity persists, for the class of circumstantial physical conditions under which classical mechanics is valid. In fact, without the existence of such a conceptual correspondence, as a *similarity between intensionally distinct but extensionally equivalent concepts*, it would seem to be impossible to identify quantum and classical mechanics as both dealing with the same world of phenomena. This is something that an insistence on radical change, without further semantic distinctions, misses.

As I said above, conceptual correspondence is not synonymous with conceptual identity: conceptual identity is intensional, and therefore it is usually not available and not needed. To be scientific realists about extensions, it is sufficient that there is a strong *similarity* between the old and the new concepts, which play the *same local roles*, so that one can in effect be reduced to the other in the appropriate domain of application. Partly through the local roles of similar concepts, we are able to identify extensions.

Another example is the electron in the hydrogen atom. The correspondence between quantum and classical mechanics usually requires, as I mentioned above, setting some quantity to a large value, compared to Planck's constant $\hbar$, i.e. doing an **approximation** or taking a **limit**. For the hydrogen atom, Bohr's correspondence principle entailed taking the limit of orbits with large quantum numbers (in units of $\hbar$); in that limit, the classical orbital frequency of the valence electron coincides numerically with the frequency of the emitted radiation, and the electron can be treated as a charged oscillating classical source. Again, there is no intensional equivalence, but there is an extensional equivalence of the quantum and the classical system, in the limit considered. That is, the electron of the Bohr model of the atom is, *pace* Kuhn and Feyerabend, *extensionally equivalent* with the electron of the classical theory of radiation: the physical systems, and the possible experimental situations, to which these models can be applied are the same, even though intensionally they are of course very different.⁵²

Also, the electron is ascribed **properties** that, while having different intensions, are, using the bridge laws, i.e. empirical identifications on the given domain of application, extensionally equivalent.⁵³ For example, the orbital frequency in one theory is identified, using Bohr's bridge law, with the frequency of the radiation that it emits in the other.⁵⁴

⁵¹Note that this correspondence goes in the opposite direction to the familiar correspondence that assigns quantum operators to classical quantities, because it attempts to give a rule for the quantum theory starting from the classical theory, rather than the other way around: see Rynasiewicz (2005). Messiah (1969: p. 70) is a *locus classicus*.

⁵²For a discussion of this case, see also Hoefer and Martí (2020: pp. 8-9).

⁵³For a list of "stable properties of the electron that are retained" by all theories after Thomson's discovery of the electron, see Bain and Norton (2001: pp. 453-455). I would like to restate Bain and Norton's claim by saying that the properties they list are intensionally different from each other, but extensionally equivalent in the sorts of experimental situations that they discuss.

⁵⁴The definition of extensional equivalence of properties is as in Section 3.1. See also Carnap (1947: p. 14), who defines both extensional and intensional equivalence of properties (as special cases

The electron of the classical and of the quantum theory, for the hydrogen atom under specified conditions, fulfils the same *causal roles* in both cases, because in both cases it interacts with the (observed) radiation in ways that are also equivalent between the two theories.

My argument, about ‘having the same local theoretical role’, for the extensional equivalence of the notion of ‘position’ of the classical and quantum theory, and of the electron of the classical and of the quantum theory, for the hydrogen atom under specified conditions, is similar to Lewis’ (1972: pp. 249-252) argument for the identification of what he calls ‘occupants of the same causal roles’, i.e. if x and y occupy the same causal role (in two different theories, viz. T and T' , respectively), and provided the occupants are uniquely realised, then also $x = y$ (where the equality sign indicates the extensional not intensional equivalence: see Carnap (1947: p. 14)).⁵⁵

The approximations or limits mentioned above are part of the level of abstraction, which determines the circumstantial physical conditions of the domain.

4.3 Material and predictive correspondence

In this Section, I discuss material correspondence and its relation with predictive correspondence, i.e. how some of the practical and descriptive aspects of reference interact with each other in an example. Material correspondence will be my main example of how the *context of application* (see Section 3.2), here given by the concrete experimental set-up, contributes to securing extensional equivalence.

Consider the case of fullerenes, which are (possibly large) molecules consisting of carbon atoms connected by single and double bonds. They can form closed meshes, with rings of five to seven atoms. A famous example is the molecule buckminsterfullerene, or C_{60} , a type of fullerene with the colloquial name “buckyball”, because its structure of twenty hexagons and twelve pentagons resembles a football. Using spectroscopy on a carbon gas obtained from e.g. a condensate of vaporised carbon, discrete infrared bands of fullerenes can be distinguished. But fullerenes can be extracted from soot in other ways, for example using organic solvents. Also, their (slow) reactions with other chemical compounds can be studied.⁵⁶

Is there an extensional equivalence or *inequivalence* between (i) the fullerene of the spectroscopist, who thinks of the substance as a quantum molecule with a discrete spectrum, (ii) the fullerene of the organic chemist, who applies solvents to the soot in a large diesel truck’s fumes to extract the fullerene molecules, and (iii) the fullerene of the chemist who is interested in how fullerene interacts with for example hydrogen or nitrous oxide gases? I claim that one can identify here, at the very least, an extensional *token identity*, of the type that Fodor (1974) discusses, that brings out the importance of **identity of**

of what he calls ‘designators’, see Section 3.1) in terms of the truth of the corresponding extensional and intensional sentences.

⁵⁵For a detailed discussion of this argument, see Butterfield and Gomes (2020). According to these authors, functionalism is a species of reduction, on Lewis’ construal of ‘reduction’, proposed as an improvement of the standard Nagelian notion.

⁵⁶See Kroto et al. (1985, 1996).

bits of matter stuff, e.g. a token sample of fullerene, in specific laboratory situations.⁵⁷ (There *may*, in addition, also be a type identity with a narrow type: but in this Section we are first interested in the role of token identity.) To see the token identity, notice that the fullerene molecules in the sample that are studied by the quantum spectroscopist are (can be) the *same* fullerene molecules in the sample studied by the chemist.⁵⁸ The identity of the tokens is easily established: simply bring a sample of fullerene, extracted by the organic chemists, to be analysed in the spectroscopy lab. (The claim here is that the extensional identity is open to explicit investigation using token identity: but it is of course not necessary to have token identity every time. A claim of equivalence also needs to be reproducible by different samples of the same type, and so only the weaker identity is needed.) Thus there is an identity of tokens, even though the intensions are different, i.e. the quantum mechanical descriptions and the classical (chemical, and material-science) descriptions are very different.⁵⁹

But notice that it is not just the identity of the bits of matter stuff that is required for full extensional equivalence: for if the chemical and the quantum mechanical descriptions did ascribe different, indeed empirically conflicting, properties to the fullerene sample, we would not think that there is equivalence. Thus *predictive correspondence* is required, i.e. the two theories should agree where their domains overlap, to the accuracy with which the domain is given by the level of abstraction. The predictive correspondence entails, in this case, numerical equivalence to the given accuracy, as well as formal correspondence. In some cases, this may be difficult to check in practice, but the numerous experiments confirmed that the atomic structure of buckminsterfullerene and other fullerenes lead to the chemical properties that the theory predicts.

Consider Feyerabend's (1963: pp. 928-932) treatment of special relativistic versus classical mass. Feyerabend admits that, in the limit of small velocities compared to the speed of light, the two masses coincide *numerically*. But he claims that their meanings disagree, because the laws in which they figure are formally and conceptually different (for example, according to Feyerabend, mass is conserved in classical mechanics, but not in special relativity).⁶⁰ I agree with Feyerabend that the intensions of the property 'mass' are different:

⁵⁷It should be clear that there is an important element of indexicality ('*this* is the sample of fullerene that we extracted from the soot') in establishing the identity of the bits of matter stuff. Hence the claim, from the previous Section, that—even though scientific theories aim to abstract from context—*context* is important when it comes to the experimental practices that enter into establishing extensional equivalence.

⁵⁸This particular argument is similar to Hacking (1983: p. 271), when he argues that people came to believe in the existence of electrons when electrons could be manipulated. However, manipulation is for him a main factor of reference, while for me it is but *one* of the factors that here contribute to establishing extensional equivalence, since I use a mixed account of reference. For a critical discussion of whether manipulability is sufficient to establish the reality of an entity, see Egg (2018: pp. 123-125).

⁵⁹Notice that the argument also does not rely on Fodor's (1974) rejection of type identity. In the present case, *even if* one granted that all organic fullerene samples are also spectroscopic fullerene samples and vice-versa, the intensions would still differ, but there would still be equivalence of extension. Thus it is equivalence of extension, rather than mere token identity, that matters.

⁶⁰In special relativity, the physical quantity of interest is the relativistic energy (related to what is sometimes called "relativistic mass" by a constant), and there is a conservation theorem for this quantity, which takes the form of the 'energy-work theorem', just as in classical mechanics. This conservation law reproduces the usual conservation law in the limit of small velocities. Thus there is not only a formal,

but once again, the extensions are the same, given the circumstantial physical condition of small velocities at this level of abstraction.⁶¹

This is because of a combination of material, formal, and conceptual correspondence. First, the mass is measured using the **same procedure** in both theories: for example, the mass of the electron was measured historically through a combination of cathode tube and oil drop experiments (see e.g. Bain and Norton (2001: p. 453)). Thus a *single sequence or combination of experiments* suffices to determine the mass in both theories,⁶² because of the material continuity between them, without which mere numerical agreement would be insufficient—which again highlights the importance of the context of application in establishing extensional equivalence.⁶³ In other words, without identity of the bits of matter stuff one would not know which numbers one should measure.

Second, in the limit in which the two theories give the same value for the mass, the mass stands in the *same relations* to other quantities in both theories. For example, in the case of the measurement of the charge of the electron: the electric field, the deflection of the trajectory of the electron, and the electric charge. This conceptual and formal correspondence secures that the material continuity relates the same quantities measured: since the theory enters in combining two experiments to “measure” the mass (see footnote 62). In other words, if the formal and conceptual correspondence of the two masses were false, then one should, despite the numerical agreement, have *denied* that a single sequence of measurements suffices to measure the quantities in the two theories. Thus *material, formal, and conceptual correspondence* work together, as one expects in a mixed account of reference, to secure that mass in special relativity and in classical mechanics are extensionally equivalent.

but also a conceptual correspondence between special relativity and classical mechanics.

⁶¹Galison’s (1997: p. 812) historical analysis of the measurement of the “relativistic mass” of the electron (or, rather, of what was then called the ‘transverse mass’), and of Einstein, Lorentz, Poincaré, and Abraham’s engagement with it, shows quite convincingly that all of these theorists are able to engage with the same experiment, communicate with each other about it, and even change their minds in the face of it. For example, Lorentz dramatically declared: ‘my hypothesis [of explaining mass by] the flattening of electrons is in contradiction with Kaufmann’s results, and I must abandon it. I am, therefore, at the end of my Latin’ (p. 812). And Poincaré made similar concessions. Galison goes on: ‘despite the “global” differences in the way “mass” classifies phenomena in the Lorentzian, Abrahamian, and Einsteinian theories, there remains a localized zone of activity in which a restricted set of actions and beliefs is deployed. In Kaufmann’s and Bucherer’s laboratories, in the arena of photographic plates, copper tubes, and electric fields, and in the capacity of hot wires to emit electrons, experimenters and theorists worked out an effective, though limited, coordination between beliefs and actions. What they worked out is, emphatically, not a protocol language—there is far too much theory woven into the joint experimental-theoretical action for that. Second, there is nothing universal in the establishment of jointly accepted procedures and arguments’ (p. 813).

⁶²This episode also highlights the combination of theory and experiment involved in the mass “measurement”, however this is not the point that I wish to stress here.

⁶³This way of establishing the token identity of bits of matter stuff echoes Galison’s (1997: p. 804) emphasis of the importance of laboratory practice and material culture: ‘different global meanings can nonetheless come to (even very complex) coordination in specific contexts. Such a partial sharing of meanings became salient at many points in the history of laboratory practice and the material culture of physics’. See also Radder’s (2012) [1984] idea of material realisation.

4.4 Overview of the elements of correspondence

I have gone through a number of examples, highlighting descriptive and practical aspects of correspondence that are jointly surely sufficient to establish extensional equivalence. I have argued that extensional equivalence requires *both* descriptive and practical aspects, as in a mixed account of reference, and as I already announced in Section 2.3.

As we have seen in our three main examples—the correspondence between quantum and classical mechanics, the case of fullerenes, and the extensional equivalence of mass between special relativity and classical mechanics—*predictive, conceptual, and material correspondence* are inter-dependent. All three are in general required to establish extensional equivalence.

In general, extensional equivalence also requires *bridge laws*, which say how the terms in one theory are to be related to those in the other. In all the cases we have seen, there are *numerical and formal correspondences* between theories. *Numerical identity* holds in an approximation that is appropriate to the particular extension, and is part of the empirical adequacy of the theories. Underlying these identities is a *conceptual correspondence*, namely a strong conceptual similarity between the concepts of the theories, so that the concepts have the same local roles in the given extension (for some concepts, one may have a stronger conceptual *identity* on a given extension). For example, two entities can have different *properties*, but these properties have the same instantiations on the relevant extension. Also the *causal roles* of entities, and more generally their *relations* to other entities, can be equivalent on an extension. Finally, conceptual correspondence includes agreement in other *possible, counterfactual situations*.

Note that the equivalence of the extensions of the terms of different theories does not only hold for obviously observational terms; it also holds for theoretical terms, such as ‘state’ and ‘aether’, that make an empirical difference.

5 A First Comparison with Other Views

As I already mentioned in Section 2.2, my mixed account of reference builds on Lewis (1984) and especially Psillos (2012). But most of all, the account should be seen as a realist improvement of, and it was directly inspired by, Nagel (1961, 1979) and Schleffler (1967: pp. 45-65): both of which include replies to Feyerabend.

I also discussed how my use of correspondence relies on Post (1971), Hartmann (2002), and Radder (1991): especially the distinction between predictive and conceptual correspondence. And the *practical* aspect of extensional equivalence that I have discussed the most, namely material correspondence, of course builds on Hacking (1983) and on Radder’s (2012) [1984] idea of material realisation.

I shall discuss the differences between extensional scientific realism and structural realism in (Part II). In Section 5.1, I compare with further aspects of Radder (2012) [1984] and further discuss the simple view of reference. In Section 5.2, I address another criticism by Stanford.

5.1 Extensional equivalence, co-reference, and the simple view

The major difference of my view with Radder (2012) [1984] is that extensional scientific realism uses a mixed account of reference, building on Lewis (1984) and especially Psillos (2012) (for some differences with these authors, see footnote 15 and the end of Section 3.2). By contrast, Radder (2012) gives a *causal* account, where the central notion of ‘material realisation’ is taken to be a *sufficient condition* for reference (p. 94): whereas in the mixed account, a practical aspect (in a broad sense, not necessarily an actual material realisation) is only *necessary*.⁶⁴

Further, Radder (2012) is adamant about the *absence of conceptual correspondence* between some of the main theories at issue in the debate over scientific realism, notably between classical and quantum mechanics: ‘An important result of analyses of the actual historical development of the natural sciences is the insight that there have been big and radical conceptual discontinuities’ (p. 83).⁶⁵ This echoes Stanford’s strong language, quoted in Section 2.2.

Indeed, Radder states in general that there is no conceptual correspondence between quantum and classical mechanics: ‘In general the correspondence is not of a conceptual but rather of a formal and numerical nature’ (p. 214). But of course I argued, in Section 4.2, against this: namely, extensional equivalence applies not only to the formal, but also to the conceptual aspects of scientific theories, including the case of classical and quantum mechanics. And so, one is justified in being a realist about these.

This shows that my notion of ‘extensional equivalence’, despite the similarities, is very different from Radder’s (2012: p. 93-105) construal of ‘co-reference’. For Radder (2012: p. 95), although the *terms* of theories refer, the *theory*’s role in co-reference is limited to giving an experimental description: rather than saying, as I do, that the conceptual correspondence of the theoretical descriptions is a necessary condition for the extensional equivalence of two theories. Indeed, for Radder, co-reference is given only by the material realisation, and the possibility to reproduce it in the future: theoretical continuity is not necessary (p. 96). But, as I argued in Section 4.3, practical continuity is not enough, since extensional equivalence in general also requires predictive and conceptual correspondence.⁶⁶

⁶⁴In so far as Radder (1991, 2012) also gives some value to predictive correspondence, he vacillates between entity realism and structural realism. However, Radder (2012), at p. 182, is clear in ultimately *rejecting* structural realism: ‘formal correspondences serve to identify potentially coreferring terms... these correspondences are not themselves interpreted in a realist way’.

⁶⁵Radder (p. 89) of course does not deny, as indeed no one does, that there are also some continuities.

⁶⁶Furthermore, Radder (2012: p. 94) requires that, to establish co-reference, experiments should be reproducible *by lay people*, who receive instructions in a ‘language of the material realisation’, i.e. a subset of ordinary language through which lay people are told how they should carry out the experiment. In view of the complexity of e.g. particle accelerator experiments, which are collaborations sometimes involving thousands of highly specialised scientists, engineers, computer scientists, and technicians, this is surely an unrealistic (and superfluous) requirement. In the mixed theory of reference, the description of the practical elements of reference is carried out by the theory itself, and the extensional equivalence of those elements and practices is described by the extensional equivalence of the descriptions. The relevant language is the language of (experimental) physicists (cf. the end of Section 3.1): and not some translation into layman terms. Recall Galison’s pidgins: cf. footnote 32. For my own response to questions about ‘theory-ladenness’, see De Haro (2019: pp. 5-6; 2020: pp. 22-23).

Ultimately, our differences—i.e. the reason why Radder judges, incorrectly, that there is no conceptual continuity between theories—seems to stem from his apparently implicitly assumed alignment of the ‘conceptual’, ‘theoretical’, and ‘intensional’ aspects (see e.g. (2012: p. 85)). Namely, *co-reference* is claimed on the sole ground of material continuity, while conceptual continuity is explicitly *denied*. I have argued that this is incorrect: for the conceptual and theoretical comprise *both* extensions and intensions.

This mix-up of ‘concept’ and ‘intension’ appears to go back to Fine (1975: p. 19), who declares: ‘I use *concept* here indistinguishability from ‘sense’ or ‘intension’ and *reference* indistinguishability from ‘denotation’, ‘extension’, or ‘referent’. In general, I shall be careless about the niceties of philosophical semantics’.⁶⁷ But this is a form of the ‘simple view’ about reference that I discussed in Section 2.2. Indeed, Fine’s self-declared ‘carelessness’ introduces a confusion between concepts and intensions (and between ‘reference’ and ‘referent’!).

It is worth briefly comparing my notion of ‘extensional equivalence’ to Kripke’s (1972: p. 99) ‘theoretical identifications’. As is well-known, Kripke believes that science is in the business of making theoretical identifications that he argues are necessary, but *a posteriori*. The following are two of his examples:⁶⁸

$$\text{water} = \text{H}_2\text{O} ; \tag{1}$$

$$\text{heat} = \text{molecular motion} . \tag{2}$$

These identities are discovered empirically, but they are true in every possible world where the terms apply.

Here, it is important that ‘water’ and ‘heat’ are *rigid designators*: ‘water’ refers to whatever is the substance that we associate with having the characteristic feel, appearance, and taste of water in the actual world (Kripke, 1972: p. 128). Likewise, ‘heat’ refers to a phenomenon that causes the characteristic sensation which we call the sensation of heat (p. 129). The latter identifications are not necessary, since *we* might have not existed (Kripke calls them *a priori*, by which he means that it can be known by philosophical analysis of the terms).

When considering theoretical identifications such as Eqs. (1)-(2), one needs to distinguish two questions: (1) The metaphysical question, independent of natural science, of whether this kind of identity is always a necessary identity, i.e. whether when two ‘rigid designators’ are identical, they are *necessarily* identical. (2) The epistemic question of whether *natural science* in fact discovers such theoretical identifications empirically.

My target here will be (2), independent of the answer to (1). The extensional realist view that I defend, while not denying that the discovery of necessary identities such as those in Eqs. (1)-(2) might in principle be possible, objects that all examples of similar identifications in empirical science that we know of, including Kripke’s own, are *contingent* rather than necessary. For example, the identity in Eq. (1) only holds between temperatures of approximately 273.2 – 373.2 K, at standard pressure, and so it is not

⁶⁷In its turn, this seems to be based on Putnam’s (1973, 1975) causal theory of reference and on Feyerabend’s (1963) insistence on meaning change. See footnote 13.

⁶⁸See Putnam (1975: p. 223-234) for further details about the water example.

in general true. The specification of these conditions is of course (part of) what I have called the specification of ‘circumstantial physical conditions’. Only then is H_2O in its liquid state: outside of this range of values, H_2O is either ice or vapour, but not water. Likewise, the controlled translational, collisionless motion of a single molecule is not heat: heat only results when the kinetic energy is transferred (typically, to the walls of a container) in collisions. Thus Eq. (2) is not true in general either, but only when appropriate circumstantial physical conditions are specified.⁶⁹

Thus in both cases one can find simple situations where Eqs. (1)-(2) are *not* identities. They are identities only if particular *states* of the world are specified. In other words, Eqs. (1)-(2) are not true in every possible world where the terms apply. Thus they are *contingent* statements, indeed cases of extensional equivalence.

Notice that the objection here is not that ‘Kripke does not know physics’ (which I believe he does); or, to put it more mildly, it is not the epistemic objection that ‘we might have been wrong about the nature of water or about the nature of heat, and so we might need to revise Eqs. (1)-(2)’. For any revision of Eqs. (1)-(2) will itself require a specification of circumstantial physical conditions and context of application. Thus the objection is that, on an extensional realist view, statements of the kind Eqs. (1)-(2) are *contingent*: in order for them to be true, in general they require a specification of circumstantial physical conditions and context of application.⁷⁰ Science does discover approximate *truths*, but *contingent* ones.

5.2 A Pyrrhic victory? Stanford’s historicism

In this Section, I address an important criticism of the recent scientific realist views discussed in Section 2.2: namely, Stanford’s (2006: pp. 144-180) charge that *selective confirmation versions of scientific realism* require either knowing facts about the beliefs and intentions of past scientists that we do not have access to, or require us to say that past scientists have repeatedly misidentified the parts, features, or aspects of their theories that genuinely contributed to their success (p. 168). At its worst, the charge is that some defenders of selective confirmation incur in a selective reading of the historical record (p. 174). That is, at least some forms of selective realism seem liable to the accusation of doing Whig history. Stanford calls this scientific realism’s ‘Pyrrhic victory’.

So the question for me is: is my extensional scientific realism vulnerable to the same accusation?

The answer is a clear ‘No’, as I will now substantiate. First, one main common feature of some of the selective versions of scientific realism that Stanford discusses, and that puts them potentially in conflict with the historical record, is the fact that they distinguish between different ‘parts, features, or aspects’ of theories, and then ascribe different “weights” to these parts. For example, some parts may be more causally involved than others in producing the relevant phenomena; or some parts may incite stronger belief

⁶⁹One might object that ‘Kripke does not mean to take identities like Eqs. (1)-(2) literally’, but rather that he allows some asymmetry in his use of the word ‘identity’. But it is clear, from his general discussion of the word ‘identity’ on pp. 106-109 (including the use of the equality sign), and the discussion of necessity that follows, that he really does mean ‘identity’, including its property of symmetry.

⁷⁰A unified theory, if such a theory is forthcoming, is perhaps an exception.

than others. Those parts that are identified as more important, significant, or sound are then, *ex hypothesi*, expected to recur as parts of *later theories*: thus the historical record should preserve them, and in this way continuity is secured. Although this commonsensical view about scientific theories may at first sight strike one as sensible, one can see how a philosophical theory based on it can easily drift into a historical claim about how theories succeed one another.

In addition to the *historical* claim, this form of realism seems to also require substantive *scientific* claims about the relative merits of different theory parts. Those claims are no doubt made partly through a careful study of the historical record, and partly using the vantage point of *current theory*. In which case, one is liable to Stanford's critique.

But extensional scientific realism is *not a form of selective confirmation*, in Stanford's sense.⁷¹ For it does not distinguish between different theory "parts" of the kind just discussed. Rather, it is a *linguistic and philosophical theory* about how to interpret the statements of scientific theories. It treats the different sentences, individual expressions (terms) and predicators of a theory *uniformly*, i.e. it does not give more weight to one type of sentence than another, one term than another, or one example of a predicator than another. It also does not in general distinguish occasions of utterance (other than in terms of the relevant context of application), i.e. the terms are used homogeneously. This is why we never needed to decide which terms refer and which ones do not: for all we know (assuming empirically adequate theories and sound scientific arguments: cf. the previous Section), *all the terms that scientists claim refer, do refer*. Rather, the main question was to establish that there is an extensional equivalence between (the terms used by) well-confirmed theories that succeed one another on a given domain of application, regardless of whether the terms have the same, or different, names.

This is worth stressing: that whenever scientists have good reasons to introduce entities into a theory, and in so far as the resulting theory is empirically supported, extensional, though in general not intensional, belief in those entities is justified.

Some amount of hindsight if of course inevitable in *any* discussion of the pessimistic meta-induction argument, since the point at issue is precisely the truth of *past* theories. For example, delimiting, in precise terms, the domain of application where a theory is well-confirmed is something that can, partly, be done only with *some* hindsight (and not necessarily the hindsight of *current* theory!): i.e. by finding domains where the theory fails to be confirmed, or by contrast with another theory that immediately succeeds it. Nevertheless, there is genuine reference already when the theory is formulated: when Newton says that there is a force of gravity whose properties are such and such, he secures reference in a domain of application whose boundaries are already largely established (e.g. the motions of the planets to the accuracy admitted at the time), although perhaps not yet very precisely. Namely, he secures reference in the domain of application to which he has *actually successfully applied* his theory. Thus, as long as scientists restrict

⁷¹Stanford (2006: p. 159) defines 'selective confirmation' as the view according to which 'those parts or aspects of rejected theories actually responsible for their successes have been preserved in or ratified by subsequent theorizing about nature, and these were the only components of past theories... for which we had any real confirmation in the first place'. Extensional scientific realism does not fall under this definition, since it does not distinguish *theory parts*, but rather distinguishes two different kinds of *meanings*.

their claims to the domain of application for which they have evidence, this not require hindsight in any negative sense.

A further question is then whether, and how, this domain of application can be further extended: to allow greater accuracy, to describe other cosmological situations, etc. It is of course further experimental work, and the contrast with later theories, that further determines, and expands, the boundaries of the domain in which belief in this referent is justified. But Newton’s original evidence already secures genuine reference in a significant domain of application: and, in so far as the theory is well-tested there, he is also justified in his belief in the theory’s posits.

This is a benign form of hindsight, that is part of normal theory construction and does not require Whig history. For, as I discussed in Section 2.3, scientists normally will not make claims (other than as their educated guesses or expressed *expectations*) about the empirical adequacy of their theories *beyond* the domain of application where the theory is well-tested: namely, a *claim* that a theory is valid beyond the domain where it has been well-tested would be—to use a Popperian word—dogmatic, and lacking in argumentative power (cf. my ‘good reasons’, above). Such claims are usually not made as part of the work of scientists.

Thus we do not need to change the meanings of the words of scientists, since they are limited to particular extensions. Rather, extensional scientific realism distinguishes two *kinds* of meaning, viz. one extensional and one intensional. It also explains why we are justified in believing in the reference of each term as regards its extension but not in general its intension. Thus the theory makes no *specific* historical or scientific claims about the pronouncements of scientists, over and above the ordinary analyses in history and philosophy of science.⁷² Rather, it *adds* an extra philosophical layer to the interpretation of the statements made by scientists. It says *in which manner the statements ought to be interpreted and believed*, i.e. about their meaning: without changing those statements themselves. This is one of the things that philosophy is supposed to do, and regularly does.

Thus my proposal is analogous to what van Fraassen (1980: pp. 14, 43, 57) does when he tells us that both realists and empiricists should listen to what scientists say, i.e. both can agree about ‘the picture of the world drawn by the theory’, but the realist and the empiricist have different degrees of belief in the entities that scientists postulate.

Likewise, extensional scientific realism tells us to *take literally what the scientist says, but to only believe it as an extensional not an intensional kind of meaning*.

6 Conclusion

This paper has defended a reply to the pessimistic meta-induction argument against scientific realism. I have argued (in Section 2) that the standard statement, in classical

⁷²The pessimistic meta-induction argument *itself*, as presented by Laudan (1980: p. 33), contains, in addition to a prediction about the future, a claim about the reference of certain terms in past theories. On the other hand, no one working on the pessimistic meta-induction argument can of course escape interpreting the historical record and the utterances of scientists at one point or another, and doing this partly with an eye on current theory. But this is part of normal philosophy and history of science.

referential semantics, that extensions are determined by intensions, circumstantial physical conditions and context of application, naturally suggests the use of a mixed theory of reference in philosophy of science, with both descriptive and practical aspects. Also, I have argued that a discussion of extensions in the context of the debate over scientific realism requires the explicit consideration of the theory’s domain of application. And that it is thus natural to restrict one’s realist commitments to the extensions, thus construed. The resulting position is *realist enough*, because it involves commitment to those things that the realist argues one should, with caution, be committed to: namely, unobservable entities, and conceptual and theoretical aspects of scientific theories. I also argued against the use of the anti-realist rhetoric of “great discontinuities”, which is insensitive to these sorts of distinctions. The discontinuities in the intensions are irrelevant, because an absolute commitment to them, separate from circumstances of evaluation and context of application, is both unnecessary and unjustified.

Adopting a mixed theory of reference (Section 3), I further analysed (in Section 4) a number of correspondence relations that together secure extensional equivalence. On the descriptive side, we have predictive and conceptual correspondence. On the practical side, I emphasised material correspondence, although there are other important factors, like measurement procedures and instrumentation.

Unlike earlier accounts, my explication of reference does not require anachronistic analyses of historical episodes (see Section 5), because it does not distinguish between different parts or elements of a theory, but rather adds to a scientific theory an extra philosophical dimension that is not normally considered in these debates. For, as any such account must do, extensional scientific realism is not primarily concerned with actual scientists’ beliefs or intentions on various occasions, nor does it aim to reform what they say: rather, what scientists say is to be taken literally. Thus extensional scientific realism is a recipe for a specific *philosophical* interpretation of what scientists say, and a justification of the corresponding beliefs.

The companion paper discusses approximate truth, and compares with structural realism.

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